

**SOLANGE: A Provenance-Complete, Enterprise-Architected QC·AI·HPC Orchestration
Platform for Non-Druggable NSCLC Tumor-Suppressor Mutations**

A dissertation submitted to the faculty of Claremont Graduate University
in partial fulfillment of the requirements for the degree of Doctor of Philosophy

by

Doron Cohen

Claremont Graduate University

2026

Approval of the Dissertation Committee

This dissertation has been duly read, reviewed, and critiqued by the Committee listed below, which hereby approves the manuscript of Doron Cohen as fulfilling the scope and quality requirements for meriting the degree of Doctor of Philosophy.

Prof. Itamar Shabtai, Chair

Claremont Graduate University

[Committee Member]

[Committee Member]

Abstract of the Dissertation

SOLANGE: A Provenance-Complete, Enterprise-Architected QC·AI·HPC Orchestration Platform for Non-Druggable NSCLC Tumor-Suppressor Mutations

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Doron Cohen

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Approximately 60% of non-small cell lung cancer (NSCLC) patients harbor mutations for which no approved targeted therapy exists. Among the most clinically significant are structural TP53 DNA-interface mutants — particularly p.Cys275Phe (C275F) — which disrupt the DNA-binding interface in ways that are intractable to classical computational drug discovery methods at the accuracy required for reliable candidate ranking. The fermionic Hamiltonian eigenvalue problem at the C275F active-space boundary belongs to a family of problems for which quantum simulation algorithms (Abrams & Lloyd, 1999; Aspuru-Guzik et al., 2005) provide efficient solutions given an adequate initial state, placing it within BQP under the standard complexity-theoretic framing of quantum chemistry. This is an operational BQP-relevance claim, not a formal separation: ground-state finding is QMA-hard in the general case, and BQP containment is conditional on sufficient initial-state overlap (§1.2). Under that framing the dual operational condition holds that (i) the best known classical methods do not achieve the required accuracy at this scale within tractable resources, and (ii) the problem admits a known BQP-class algorithmic solution in the selected operational regime.

This dissertation proposes, designs, implements, and evaluates an orchestration platform — hardware-agnostic at the workflow level; vendor-specific at the compilation and error-mitigation layer — that integrates quantum computing, high-performance classical computing, and artificial intelligence into a unified, reproducible, and FDA 21 CFR Part 11-compliant workflow for simulating non-druggable NSCLC tumor suppressor mutation targets. The primary

research artifact is a working prototype platform grounded in real patient NGS data, validated through proof-of-concept execution on a quantum simulator backend.

Three original IT contributions are claimed: a BQP-relevant classification framework for non-druggable NSCLC targets, a Part-11-governed QC·AI·HPC orchestration architecture for oncology mutation simulation, and the first compliance framework (as of June 2026) applying FDA 21 CFR Part 11 to probabilistic quantum-computed molecular binding data — including ML-decoder provenance (P9) for emerging Ising-class learned QEC. The contribution of this dissertation is the architecture — the system that makes quantum simulation of non-druggable tumor suppressor mutations accessible, reproducible, and governed. Full-scale execution on IBM Heron r3 quantum hardware is identified as a subsequent research phase, enabled by the architecture designed here.

Current targeted therapies, even where they exist, confront an additional structural barrier: de novo resistance. For the majority of NSCLC patients in the non-druggable class represented by Table 2, no first-line targeted agent has ever been identified — and therefore no resistance trajectory has even begun. HIV provides the closest historical analogue at scale: a uniformly fatal disease with no therapeutic entry point, transformed by computational structural biology and iterative molecular simulation into a manageable chronic condition. The transformation did not happen because a single drug was found — it happened because the information systems infrastructure for drug-discovery iteration was built: governed, reproducible, and grounded in molecular simulation. This dissertation proposes quantum simulation as the foundational computational step toward an analogous transformation for non-druggable NSCLC tumor-suppressor mutations — not as a therapeutic claim, but as the information systems architecture that makes iterative computational drug discovery possible for the first time in this class.

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§ 01 Problem Statement — Three Gaps, One System.

1.1 • The Clinical Gap

NSCLC is the leading cause of cancer mortality globally, accounting for approximately 2.1 million new diagnoses per year and 1.53 million deaths annually (GLOBOCAN 2022). Approximately 70% of patients are diagnosed at advanced stage (locally advanced or metastatic), when curative resection is no longer possible. Median overall survival at advanced stage ranges from 8 to 14 months, and five-year survival is approximately 35% for locally advanced disease and 7% for metastatic disease — a figure itself inflated by the subset of patients who carry actionable driver mutations (EGFR, ALK, KRAS G12C) and benefit from approved targeted therapies. For the majority who do not carry actionable mutations, median survival and five-year rates are substantially lower. Advances in targeted therapy have transformed outcomes for patients with those actionable drivers — but they represent a minority. The majority of NSCLC patients carry mutations that current drug design approaches cannot address. They have no approved targeted therapy.

Table 1 — Pipeline-Eligible Non-druggable NSCLC Mutations

Fifteen per-variant classification entries across the nine tumor-suppressor genes with complete active-space data, each satisfying the BQP-relevant classification criteria: no approved targeted therapy, fermionic active space within quantum feasibility window, and no tractable classical solution at chemical accuracy. The full pipeline-eligible set spans eleven genes; the two additional genes — SETD2 and U2AF1 — appear in the patient-burden analysis of Table 2 and await full active-space classification.

Gene	Mutation	Mutation Type	Protein Location	Active Electrons	Qubits (JW)	Primary Cancer Types	NSCLC Frequency	Key Data Sources
TP53	C275F	Structural (hydrophobic core)	DNA-binding domain, loop-sheet-helix	~44	~88	NSCLC, breast, colorectal †	<1% (anchor case)	COSMIC, IARC TP53 DB, PDB 2OCJ
TP53	R175H	Structural (global destabilization)	DNA-binding domain, β -sandwich core	~44	~88	NSCLC, ovarian, colorectal	~3–4%	Joerger & Fersht (2007), TCGA
TP53	G245S	Structural (L3 loop)	DNA-binding domain, L3 loop	~40	~80	NSCLC, head & neck	~1–2%	IARC TP53 DB, COSMIC
TP53	R249S	Structural (L3 loop)	DNA-binding domain, L3 loop	~40	~80	Hepatocellular, NSCLC	<1%	COSMIC, IARC TP53 DB
TP53	R282W	Structural (H2 helix)	DNA-binding domain, H2 helix	~42	~84	NSCLC, breast	~1%	Joerger & Fersht (2007), TCGA
KEAP1	G333C	LOF (Kelch domain)	Kelch β -propeller, NRF2 interface	~28	~56	NSCLC (adenocarcinoma)	~2–3%	TCGA, COSMIC, PDB 2FLU
KEAP1	R320Q	LOF (Kelch domain)	Kelch β -propeller, NRF2 interface	~28	~56	NSCLC (adenocarcinoma)	~1–2%	TCGA, COSMIC
CDKN2A	p16 LOF	LOF (deletion/inactivation)	CDK4/6 inhibitor interface	~24	~48	NSCLC, melanoma, pancreatic	~15–20%	TCGA, PDB 2A5E
STK11	F354L	Kinase domain LOF	Kinase catalytic domain	~26	~52	NSCLC (adenocarcinoma)	~2–3%	TCGA, PDB 2QK7
STK11	D194N	Kinase domain LOF	Kinase catalytic domain	~26	~52	NSCLC (adenocarcinoma)	~1–2%	TCGA, COSMIC
NF1	LOF	LOF (truncating)	GTPase-activating domain	~30	~60	NSCLC, glioma	~5–7%	TCGA, COSMIC
RB1	LOF	LOF (deletion/truncating)	Pocket domain (E2F interface)	~32	~64	NSCLC (SCLC), retinoblastoma †	~3–5%	TCGA, COSMIC
SMARCA4	LOF	LOF (ATPase domain)	SWI/SNF ATPase domain	~28	~56	NSCLC, ovarian, medulloblastoma	~5–8%	TCGA, COSMIC
RBM10	LOF	LOF (splicing regulator)	RNA-binding domain	~22	~44	NSCLC (adenocarcinoma)	~5–7%	TCGA, COSMIC
ARID1A	LOF	LOF (chromatin remodeling)	ARID DNA-binding domain	~24	~48	NSCLC, endometrial, gastric †	~4–6%	TCGA, COSMIC

† Also observed in urological malignancies (urothelial carcinoma, renal cell carcinoma); SOLANGE pipeline directly applicable without architectural modification.

Table 2 — Annual Patient Burden by Mutation: Non-druggable NSCLC in the Quantum Feasibility Window

Per-mutation patient counts derived from NSCLC frequency estimates applied to 2.12M new diagnoses/year (GLOBOCAN 2022). The naïve sum (2,877,000) is not a patient count: it double-counts patients, because a patient carrying several of these co-occurring mutations is tallied once in each mutation's row, so the sum exceeds the ~2.12M total annual NSCLC incidence. Removing that overlap — a ~55–62% reduction ($2,877,000 \times \sim 0.40 \approx 1.1\text{--}1.3\text{M}$) — yields the estimated number of unique patients carrying at least one non-druggable mutation. This ~1.1–1.3M is roughly 52–61% of all NSCLC patients, consistent with the ~60% non-druggable fraction stated in the Abstract. All mutations satisfy the three BQP-relevant classification criteria: no approved targeted therapy, fermionic active space within quantum feasibility window, and no tractable classical solution at chemical accuracy. This burden analysis spans eleven genes — the nine classified in Table 1 plus SETD2 and U2AF1, which contribute to the population estimate but await full active-space classification.

Gene / Mutation	NSCLC Frequency	Est. Patients/Year	Notes
TP53 C275F (anchor case)	~2%	~42,000	Structural mutant, loop-sheet-helix junction; no PDB entry; no clinical trial as of April 2026
TP53 R175H	~5%	~105,000	Global destabilization; most common TP53 hotspot; gain-of-function activity reported
TP53 G245S	~3%	~64,000	DNA-contact region; partial contact + structural destabilization
TP53 R249S	~2%	~42,000	L3 loop; associated with hepatocellular carcinoma; NSCLC co-driver
TP53 R282W	~2%	~42,000	Hydrophobic core; structural class; no targeted therapy
KEAP1 LOF (G333C, R320Q + others)	~17%	~360,000	Predominantly adenocarcinoma; no approved direct KEAP1 inhibitor
CDKN2A p16 LOF	~17%	~360,000	CDK4/6 inhibitors approved in breast cancer; not NSCLC-specific
STK11/LKB1 LOF	~20%	~424,000	Major driver of immunotherapy resistance; no direct STK11 therapy
NF1 LOF	~6%	~127,000	RAS-pathway; MEK inhibitors partially address; no direct NF1 therapy
RB1 LOF	~4%	~85,000	Predominantly SCLC component; no direct RB1 therapy
SMARCA4 LOF	~6%	~127,000	SWI/SNF complex; no approved targeted therapy
RBM10 LOF	~6%	~127,000	Splicing regulator; no approved targeted therapy
ARID1A LOF	~5%	~106,000	Chromatin remodeling; no approved targeted therapy
SETD2 LOF	~5%	~106,000	Histone methyltransferase; tumor suppressor; no approved targeted therapy
U2AF1 S34F/Y	~3%	~64,000	Splicing factor mutation; no approved targeted therapy
Naïve sum (pre-correction)	—	~2,877,000	Exceeds annual incidence due to co-occurring mutations in the same patient
Overlap-corrected total (~55–62% correction)	—	~1.1–1.3M unique patients/year	Unique patients carrying ≥ 1 non-druggable mutation within the quantum feasibility window

The magnitude of this unaddressed population reflects a structural limitation in the current drug-discovery pipeline: de novo resistance. Even among mutations that initially yield to targeted therapy — such as EGFR exon 19 deletions responding to osimertinib — resistance emerges within 12–18 months in the majority of patients. For the non-druggable class represented in Table 2, no first-line targeted therapy exists to trigger resistance in the first place. The pipeline has not simply failed these patients; it has never begun. A historical analogy clarifies the scale of the unsolved problem: HIV infection was once uniformly fatal within years of diagnosis, with no therapeutic options and no drug-discovery framework capable of attacking the virus's protease at the required precision. The development of combination antiretroviral therapy — grounded in computational structural biology and iterative molecular simulation — transformed HIV from a death sentence into a manageable chronic condition. The quantum simulation platform proposed in this dissertation is not a claim to replicate that transformation; it is a claim to establish the foundational information systems infrastructure — governed, reproducible, and Part-11-

compliant — that makes iterative computational drug discovery possible for the non-druggable tumor-suppressor class, in the same way that reliable viral load measurement and provenance-tracked resistance profiling made antiretroviral therapy iteration possible for HIV.

Among the most challenging are TP53 structural DNA-interface mutants. Unlike cavity-creating mutants such as Y220C, which creates a druggable pocket, structural DNA-interface mutants such as p.Cys275Phe disrupt the DNA-binding interface and impair p53 transcriptional function. The Cys-to-Phe substitution introduces a bulky aromatic side chain near the loop-sheet-helix DNA-contact region of the p53 DNA-binding domain, likely perturbing local DNA-binding geometry rather than creating a therapeutically exploitable cavity. No approved drug specifically targets TP53 C275F, and no C275F-specific clinical trial appears to be registered on ClinicalTrials.gov as of April 2026.

The clinical urgency of this problem is not hypothetical. The patient NGS data anchoring this research (report MI25-0349, Emek Medical Center, June 2025) identified TP53 p.Cys275Phe as a Tier I pathogenic variant at allele frequency 12.66% in a pulmonary adenocarcinoma case — alongside KEAP1 p.Cys196Leufs* and CDKN2A deletion, with no EGFR, ALK, KRAS, or RET detected. This patient had no approved targeted therapy available.

The structural inaccessibility of C275F compounds the clinical gap. As of April 2026, no crystallographic structure of the TP53 C275F mutant exists in the RCSB Protein Data Bank. For comparison, the Y220C mutant — which creates a druggable pocket — has been crystallised and deposited (PDB: 2VUK), enabling structure-based drug design. C275F has not: the Cys → Phe substitution destabilises the hydrophobic core without producing a tractable crystal form, and the resulting conformational heterogeneity prevents X-ray structure determination at useful resolution. The wild-type backbone shown in Figure 1.1 is therefore not a starting point for classical docking — it is the closest available approximation to an unknown structure. The absence of a C275F PDB entry is not a gap in the literature that will be filled by a better crystallography experiment; it is a signal that the mutation's structural consequences must be computed, not observed. Quantum simulation is not one option among several — it is the only method that can characterise the electronic structure of C275F at chemical accuracy, at this active-space scale, with current technology.

This dissertation is a consumer of PDB data, not a producer of it. The structural record above (PDB 2OCJ) is an input to the orchestration pipeline — parallel to the clinical NGS report MI25-0349 that enters the same pipeline at the patient data layer. The platform's first operation upon receiving a mutation target is to retrieve the relevant PDB entry, extract the coordinate data required to define the active-space boundary, and log the PDB identifier and retrieval timestamp in the provenance record. What the platform does with that input — active-space selection, Hamiltonian encoding, quantum simulation routing, error mitigation, and Part-11-compliant output generation — is the IT contribution. The structural biology is upstream of this architecture; the information systems work begins at the point of ingestion.

Why C275F and Not Another Mutation?

The selection of TP53 p.Cys275Phe (C275F) as the anchor case satisfies five independent criteria, each of which alone would justify its choice, and which jointly make it a uniquely suitable demonstration target for a quantum-simulation orchestration architecture.

Criterion 1 — Therapeutic orphan status. C275F has no approved targeted therapy and no registered clinical trial as of April 2026. The platform is designed precisely for the class of mutations where classical drug-discovery pipelines have failed. Mutations with existing therapies (e.g., TP53 Y220C with Rezatapopt in Phase II) are poor demonstration targets because the platform would be solving an already-solved problem. C275F is unambiguously in the unsolved class.

Criterion 2 — Absence of crystallographic data. No structure of the C275F mutant exists in the RCSB Protein Data Bank. The Cys → Phe substitution destabilizes the hydrophobic core of the DNA-binding domain, producing conformational heterogeneity that prevents X-ray crystallography at useful resolution. The electronic structure of C275F cannot be observed — it must be computed. This makes

quantum simulation not one option among several but the primary path to chemical-accuracy characterization, which is precisely the condition under which the BQP-relevant classification framework applies.

Criterion 3 — Active-space size in the operationally BQP-relevant range. The full binding-site model requires approximately 44 active electrons, mapping to approximately 88 qubits under Jordan-Wigner encoding. This places the problem above the classical tractability threshold for coupled-cluster methods and within the demonstrated range of the SqDRIFT framework (IBM, March 2026: 32 electrons on 100 superconducting qubits). C275F sits precisely in the regime the platform is designed to govern — neither too small for classical methods nor too large for near-term quantum hardware.

Criterion 4 — Structural mutation class strengthens the quantum argument. As established in §1.2, C275F is a structural mutant: the perturbation is delocalized across the hydrophobic core of the DNA-binding domain rather than local to a single DNA-contacting residue. This delocalization produces correlated electron behavior across multiple residues simultaneously — the regime where classical DFT fails at chemical accuracy and where fermionic quantum simulation provides a principled advantage. A contact mutant such as R248W, by contrast, involves a more localized perturbation that is in principle more tractable classically. C275F is therefore a harder problem in exactly the right way.

Criterion 5 — Clinical anchoring in real patient data. The mutation is sourced from a de-identified clinical NGS report (MI25-0349, Emek Medical Center, June 2025) identifying TP53 C275F as a Tier I pathogenic variant at 12.66% allele frequency in a pulmonary adenocarcinoma case with no actionable targeted driver detected. Using a real clinical case as the motivating instance grounds the artifact in an authentic translational problem, consistent with DSR's requirement that artifacts address real, significant organizational and technical problems rather than constructed ones.

The selection of a single mutation as the anchor case is a deliberate methodological choice, not a claim that C275F is uniquely important among all non-druggable tumor-suppressor mutations. The platform is hardware-agnostic and mutation-agnostic at the workflow layer: the same orchestration architecture, provenance schema, and compliance framework apply to any fermionic target satisfying the BQP-relevant classification criteria. C275F is the existence proof; analytical generalization to other non-druggable structural mutants follows from the architecture's design, not from additional case studies.

1.2 · The Computational Gap

Classical computational chemistry cannot reliably model the binding energetics of TP53 C275F at the required accuracy level. DFT (B3LYP) produces errors of ± 5 –10 kcal/mol against the ≤ 2 kcal/mol chemical accuracy threshold. CCSD(T) achieves gold-standard accuracy (± 1 kcal/mol) but scales as $O(N^7)$, becoming computationally infeasible at ~ 44 active electrons. Full Configuration Interaction (FCI) scales as $O(2^N)$ — reaching ~ 17 trillion operations at $N=44$, rendering it completely intractable.

The full binding pocket of TP53 C275F requires approximately 44 active electrons and 44 active orbitals. Jordan-Wigner encoding maps this to ~ 88 qubits — within the capacity of IBM Heron r3 (156 qubits). VQE with sqDRIFT Hamiltonian simulation provides polynomial circuit-depth scaling, placing this problem within BQP under the standard quantum-chemistry framing (Abrams & Lloyd, 1999; Aspuru-Guzik et al., 2005) and making the active space tractable on NISQ-era hardware.

The structural — rather than contact — nature of C275F further strengthens the quantum simulation argument on three independent grounds. First, structural mutations produce delocalized electronic perturbations across the entire active site: the Cys $\rightarrow$ Phe substitution reorganizes the hydrophobic core of the DBD, creating correlated electron behavior across multiple residues simultaneously. Classical DFT methods, which treat exchange-correlation locally, cannot capture this non-local electronic reorganization at chemical accuracy — precisely the regime where SqDRIFT's polynomial-time fermionic simulation provides advantage. Second, the absence of a C275F crystal structure in the RCSB PDB is itself a direct consequence of structural instability: conformational heterogeneity induced by the mutation prevents X-ray

structure determination at useful resolution, confirming that the electronic consequences of C275F must be computed rather than observed. This makes quantum simulation not merely advantageous but the only viable path to chemical-accuracy characterization. Third, the 44-electron active space required for Phase 3B is scientifically justified by the structural classification: unlike contact mutations — where the perturbation is local to a single DNA-contacting residue — structural mutations require inclusion of the broader electronic environment encompassing the Zn^{2+} coordination shell and adjacent hydrophobic core residues. The two-level active space framework (24 electrons local, 44 electrons full) is therefore not an arbitrary design choice but a direct consequence of C275F's structural mutation mechanism.

This dissertation does not claim a formal complexity-theoretic separation between classical and quantum solvability for this problem; such separations remain open in general. The claim is the weaker but defensible one — that C275F sits in the empirically quantum-advantaged regime as defined by the current state of both classical and quantum methods circa 2026. Advanced classical approximations are explicitly acknowledged: tensor-network methods (DMRG) are highly effective for one-dimensional and quasi-1D systems but degrade on three-dimensional, pocket-like active spaces with strong correlation across non-local orbitals; neural quantum states (Carleo & Troyer, 2017) have shown promising results on lattice models but have not yet demonstrated chemical accuracy on fermionic systems at the ~44-electron metalloprotein scale required here. The platform's BQP-relevant framework remains valid until — and only until — a classical method demonstrates chemical accuracy at this active-space boundary within tractable resources, at which point the architecture's hardware-agnostic abstraction would simply route the workload to that classical backend.

Two complexity-theoretic subtleties are stated directly, since they are easy to overstate. First, finding the ground state of a general local Hamiltonian is QMA-hard — the quantum analogue of NP-hardness — so no polynomial-time quantum algorithm is guaranteed in the worst case. The containment of a specific instance in BQP is conditional on preparing an initial state with sufficient overlap on the true ground state (the overlap problem), and constructing such a state is itself hard in general. This dissertation therefore makes no claim of guaranteed exponential quantum advantage. Second, and directly on point, Lee et al. (2023) — Evaluating the evidence for exponential quantum advantage in ground-state quantum chemistry — argue that for many ground-state chemistry problems the initial-overlap requirement and the strength of classical heuristics together erode the case for exponential speedup, concluding that evidence for exponential quantum advantage across chemical space has yet to be found. That critique is accepted here as correct in the general case; rather than undermining this work it delimits it. The claim advanced is not exponential advantage in principle but a practical quantum algorithm in the specific operational regime of the active spaces selected — where Hartree–Fock/CASSCF reference states supply adequate initial overlap and the best classical methods do not reach chemical accuracy at tractable cost — with the dissertation's contribution being the governance and orchestration architecture around that algorithm, not a proof of quantum supremacy for chemistry.

1.2.1 • Active Space Definition — Two-Level Framework

This subsection defines a system requirement for the orchestration architecture, not a chemistry claim. For the workflow to handle C275F end-to-end, it must commit to a precise active-space boundary; that boundary determines qubit count, telemetry volume, and audit-trail granularity downstream. The chemistry is anchoring detail; the IT contribution is the workflow's ability to execute reproducibly at either boundary. Two levels are defined, each corresponding to a distinct architectural validation phase.

Local Binding Site • ~24 Active Electrons

Encompasses the immediate chemical environment of the mutation: the F275 aromatic π -system (6 electrons), adjacent residues C277 and A276 (~10 electrons), and relevant bonding orbitals between them (~8 electrons). This level is sufficient for quantum-accurate drug candidate identification — characterizing the altered binding pocket geometry, evaluating candidate molecule interactions, and comparing wild-type versus mutant electronic structure. Requiring approximately 48 qubits under Jordan-Wigner encoding, this active space is directly

feasible on IBM Heron r3 (156 qubits) and is enabled by the SqDRIFT benchmark (IBM Research, 2026; arXiv:2603.08696: 32 electrons demonstrated on 100 qubits).

Full Binding Site · ~44 Active Electrons

Extends the local site to incorporate the complete biologically active environment: the Zn^{2+} coordination shell and its four ligands (C176, H179, C238, C242), plus the DNA-guanine interface and phosphate backbone (~12–20 additional electrons). This level is required for full functional validation — confirming whether a drug candidate restores p53 DNA-binding activity. Jordan-Wigner encoding maps ~44 electrons to ~88 qubits, placing this within IBM Heron r3 capacity but at the frontier of demonstrated capability.

System	Active Electrons	Qubits Required	Feasibility
C_{13}Cl_2 (IBM Research 2026 benchmark)	32	~100	✓ Demonstrated
C275F — Local binding site	~24	~48	✓ Feasible today
C275F — Full binding site	~44	~88	► Near-term frontier

1.3 · The Systems Gap

Even if quantum hardware can simulate TP53 C275F, a deeper IT problem exists: no integrated system connects the clinical problem to a quantum simulation result in a governed, reproducible, stakeholder-accessible way. Quantum platforms are vendor-specific. HPC tools run in isolated environments. AI rescoring models require separate pipelines. No orchestration layer connects them. No compliance framework governs the output. No visualization layer communicates results to the oncologists, pharma researchers, and hospital IT teams who need to act on them.

1.4 · Hardware Modality Anchor

The modality choice is an IT decision, not a physics ranking. The orchestration architecture must commit to a single hardware anchor whose telemetry is well-specified, whose calibration epochs are exposed, and whose compiler stack supports the audit-trail framework of §06.iii — otherwise the platform cannot satisfy 21 CFR Part 11 reproducibility requirements at proof-of-concept. The four-modality survey below is the engineering rationale for that commitment, not a scientific judgement about which physical substrate is "best."

The orchestration architecture is hardware-agnostic at the workflow level; vendor-specific at the compilation and error-mitigation layer. IBM Heron, IonQ, neutral-atom systems, and HPC backends are all addressable through the same workflow abstraction, but compilation strategy (SWAP overhead on superconducting; all-to-all on trapped-ion; reconfigurable geometry on neutral atoms) and error-mitigation primitives (ZNE tuned for superconducting noise; different dominant error channels elsewhere) cannot be made fully portable. The contribution is to minimize this leak through a documented translation layer, not to eliminate it. The proof-of-concept execution, however, must anchor on a single quantum modality. The four commercially accessible paradigms in 2026 — superconducting circuits, trapped ions, neutral atoms, and photonic systems — differ materially in fidelity, connectivity, gate speed, and software maturity. The choice of anchor must be derived from the specific computational demands of this dissertation's target problem (gate-based ground-state estimation for fermionic Hamiltonians at 20–32 electron active spaces, executed as sample-based quantum diagonalization — quantum sampling of the dominant electronic configurations followed by classical diagonalization over the sampled subspace), not from generic benchmarks.

Eight criteria are dispositive for the present problem class: C1 active-space tractability, C2 circuit-depth fidelity, C3 wall-clock execution speed, C4 SqDRIFT compiler support, C5 HPC integration, C6 audit-trail telemetry, C7 empirical precedent at the target scale, and C8 timeline stability through the dissertation period.

Criterion	Superconducting	Trapped Ion	Neutral Atom	Photonic
C1 · Active-space scale	Strong	Adequate	Strong	Limited
C2 · Circuit-depth fidelity	Adequate	Strong	Adequate	Limited
C3 · Gate speed / wall-clock	Strong (ns)	Limited (ms)	Limited (μ s–ms)	Strong
C4 · SqDRIFT compiler	Strong (Qiskit)	Adequate	Emerging	Limited
C5 · HPC integration	Strong	Adequate	Emerging	Limited
C6 · Audit-trail telemetry	Strong	Adequate	Emerging	Limited
C7 · Precedent at target scale	Strong (SqDRIFT 2026)	Adequate	Adequate (different alg.)	Limited
C8 · Timeline stability	Strong	Strong	Adequate	Adequate

Decision: superconducting qubits are selected as the anchoring modality. The choice is the best-supported at submission for three converging reasons. First , the SqDRIFT framework — on which this dissertation's time-compression argument rests — was demonstrated in 2026 on 100 superconducting qubits recovering the ground-state energy of a 32-electron active space of $C_{13}Cl_2$. This is the only published benchmark in the world that matches the active-space size targeted here. Second , Qiskit Runtime provides production-grade primitives for fermionic Hamiltonian construction, sample-state preparation, classical diagonalization, and per-circuit telemetry sufficient to populate the audit-trail framework specified in §06. Third , SqDRIFT is fundamentally a sampling procedure — the quantum processor draws many independent samples of the dominant electronic configurations, which a classical solver then diagonalizes — so its time-to-solution is dominated by the product of per-circuit execution time and the number of sampling circuits and shots. Nanosecond gates make this product tractable; millisecond gates would inflate it by three orders of magnitude.

The choice is anchored, not absolute. A dual-modality framing (superconducting + neutral atoms) would dilute the central thesis without empirical symmetry: no SqDRIFT-class workload has been demonstrated on neutral atoms at the relevant scale. The pipeline architecture is described abstractly enough that re-anchoring on a different modality, should the empirical landscape shift before defense, is a defined exercise in future work — not a re-derivation of the core results. Two modality-specific properties propagate forward: connectivity-induced SWAP overhead (treated as a sensitivity parameter in circuit-depth estimates) and calibration drift (treated as a mandatory metadata field in the simulation provenance record — see §06.iii).

1.4.1 · Post-April 2026 Hybrid Landscape

The modality argument above was framed against the state of the field at submission. On April 14, 2026 , Nvidia released Ising — a 35-billion-parameter vision-language model for agentic qubit calibration paired with a 3D convolutional neural network that acts as a fast pre-decoder for real-time quantum error correction, resolving the common syndrome patterns and escalating the remainder to a conventional global decoder (e.g., pyMatching), reportedly 2.5× faster and 3× more accurate than the pyMatching-only baseline — alongside the NVQLink GPU ↔ QPU interconnect and continued investment in the CUDA-Q programming model. Nvidia's stated position, confirmed by CEO Jensen Huang, is that the company will not build a quantum processing unit ; it intends to occupy the classical, AI, and HPC substrate sitting beneath any vendor's QPU. This release does not weaken the modality argument made above; it confirms its structural premise. The QC·AI·HPC stack is bifurcating into a quantum-kernel layer (QPU + algorithm) and a hybrid-substrate layer (classical pre/post-processing, AI calibration, AI-decoded QEC). IBM Heron r3 + SqDRIFT remains this dissertation's quantum-kernel anchor — and IBM is notably absent from Nvidia's announced NVQLink partner roster (Atom Computing, IonQ, IQM, Q-CTRL, Academia Sinica, EeroQ), continuing its own Quantum-Centric Supercomputing path on Qiskit — independently validated at protein scale on March 26, 2026 , when Cleveland Clinic and IBM published the first electronic-structure simulation of the 303-atom Trp-cage miniprotein via Embedded Wave Function with Sample-based Quantum Diagonalization (SQD) on Heron r2. The orchestration architecture defended in §06.ii is therefore positioned to sit across this bifurcation: treating CUDA-Q, NVQLink, and Ising as commodity substrate where they apply, and Qiskit Runtime where IBM's path is preferred — rather than committing the dissertation to either side of the emerging vendor split.



[Figure — rendered from source]

1.4.2 · Trapped-Ion Logical-Qubit Landscape

On March 26, 2026, Quantinuum reported a demonstration on its Helios trapped-ion processor of quantum computations with up to 94 logical qubits using error-detection codes (and up to 48 logical qubits using concatenated error-correction codes), achieving "beyond break-even" performance — encoded qubits producing lower error rates than unencoded physical hardware operations. This is a notable milestone for the trapped-ion modality and sharpens — but does not invalidate — the superconducting modality argument made in §1.4.

A common misconception in evaluating quantum hardware for computational chemistry is to treat coherence time as the primary figure of merit. Trapped-ion systems exhibit coherence times measured in seconds — occasionally minutes in specialized configurations — while superconducting transmon systems such as IBM's Heron operate with coherence windows of approximately 100–300 microseconds. Taken in isolation, this represents a four-orders-of-magnitude advantage for trapped ions.

The operative metric, however, is not absolute coherence but the ratio between coherence time and gate time, which determines the effective circuit depth available before decoherence dominates. Trapped-ion systems combine multi-second coherence with microsecond-scale gate operations, yielding on the order of 10^6 operations per coherence window. Superconducting systems combine ~200-microsecond coherence with nanosecond-scale gates, yielding on the order of 10^4 operations per window. By this measure, trapped ions retain a meaningful advantage for deep circuits.

This advantage does not translate uniformly across application domains. Drug-discovery workloads — particularly the simulation of mutation-specific electronic structure across libraries of candidate ligands and conformational variants — are not characterized by a single deep circuit, but by a high volume of moderate-depth circuits executed across a broad parameter space. SqDRIFT — the algorithm anchoring this dissertation's empirical claim — is a sample-based quantum diagonalization method: the quantum processor samples the high-weight electronic configurations that span the relevant subspace, and a classical processor diagonalizes the Hamiltonian projected onto those samples. Reaching a converged energy requires a large number of independent sampling circuits and shots, so its time-to-solution is determined by the product of per-circuit execution time $\times$ sample count. This is the canonical throughput-bound regime: aggregate throughput, not per-circuit depth, is the binding metric. Under this regime, the nanosecond gate times of superconducting hardware confer a decisive practical advantage despite the shorter coherence window per circuit.

A March 2026 Nature report from Quantinuum on its Helios processor provides a concrete, citable data point for this throughput argument [55]. Helios is a 98-qubit trapped-ion QCCD device using ^{137}Ba + hyperfine qubits with all-to-all connectivity, achieving single-qubit infidelity of 2.5×10^{-5} , two-qubit gate infidelity of 7.9×10^{-4} (99.92% fidelity), and SPAM infidelity of 3.3×10^{-4} — fidelities that substantially exceed those reported for IBM Heron r3, and sufficient to demonstrate random circuit sampling beyond classical simulability. These figures confirm, rather than undermine, the coherence-versus-throughput distinction drawn above: the same report measures approximately 55 ms of wall-clock time per depth-1 entangling layer across the full 98-qubit register. A SqDRIFT sampling run, whose cost scales with the product of per-circuit execution time and sample count rather than with single-shot fidelity, would accumulate this per-layer latency across the thousands of sampling circuit instances the method requires. At Heron r3's nanosecond-scale gate times, the equivalent sampling budget completes in a small fraction of the time implied by Helios's millisecond-scale layer latency, even though Helios's per-gate error rates are lower. The Quantinuum result therefore strengthens rather than threatens the case made here: it is direct experimental confirmation that high coherence and high throughput are independent axes of merit, and that workload class — not platform reputation — determines which axis dominates.

The architectural choice of IBM Heron r3 as the hardware substrate for this dissertation's QC·AI·HPC pipeline therefore reflects a workload-aligned optimization, not a claim of categorical superiority. Quantinuum's logical-qubit demonstration does not weaken the modality argument made in §1.4; it sharpens its scope. Trapped-ion platforms remain well-suited to applications demanding maximal circuit depth per execution — fault-tolerance research, logical-qubit benchmarking, and certain classes of long-time-evolution simulation — domains this dissertation explicitly does not target. Drug discovery is throughput-dominated, and superconducting hardware is presently the modality whose operational profile aligns with that demand.

1.4.3 · Microsoft–Quantinuum Fault-Tolerant Milestone

A preceding and complementary milestone to the Helios results discussed above was published by Microsoft and Quantinuum in Nature under the title "Improved quantum processor logical error rates via correction and detection" [54]. Running on Quantinuum's H2 trapped-ion processor with Microsoft's qubit-virtualization software stack, the joint team demonstrated an 800-fold reduction in logical error rate relative to the physical qubit baseline — the largest recorded gap between physical and logical error rates reported in the peer-reviewed literature at the time. The system executed 14,000 individual circuit instances without a single observed error and scaled from 4 to 12 highly reliable logical qubits using active syndrome extraction with real-time classical co-processor correction, without disturbing the quantum data during error diagnosis.

The significance for this dissertation is threefold. First, it constitutes independent peer-reviewed validation that the field has moved past the NISQ era into what Microsoft terms "resilient quantum computing" — the regime in which logical qubits outperform physical qubits, the foundational requirement for fault-tolerant quantum chemistry. Second, it demonstrates that the QEC machinery required for

Phase 3B of the SOLANGE pipeline (error mitigation on IBM Heron r3, escalating to full fault-tolerance as hardware matures) is not a distant theoretical aspiration but an experimentally demonstrated trajectory. Third, the end-to-end chemistry simulation paired with AI and HPC supercomputers in the same demonstration is structurally identical to the QC·AI·HPC orchestration architecture defended in §06 — confirming the architectural pattern independently. This milestone does not change the hardware choice: Microsoft's qubit-virtualization stack runs on Quantinuum's trapped-ion hardware, not on IBM's superconducting backend, and the throughput argument of §1.4.2 continues to apply. It does, however, materially strengthen the scientific timeline argument for Phase 3B feasibility.

1.4.4 · Neutral-Atom Landscape (Inflection)

The neutral-atom modality reached a commercial-maturity milestone on February 17, 2026, when Inflection (formerly ColdQuanta) became the first neutral-atom quantum company to list publicly, trading on the NYSE under the ticker INFQ following a business combination with Churchill Capital Corp X. The listing raised over \$550 million in gross proceeds, and Inflection's systems are already deployed with the U.S. Department of War, NASA, and the U.K. government, alongside a joint demonstration with Nvidia of a materials-science application using logical qubits [56]. This commercial trajectory does not, on its own, alter the modality argument made in §1.4; it raises the question of whether the "Emerging" ratings assigned to neutral atoms in the C4-C6 rows of the comparison table still hold, and the answer requires separating two distinct claims that are easy to conflate.

The first claim is scale. Neutral-atom arrays, built from optically trapped alkali atoms with no fabrication-induced variation between qubits, scale favorably in qubit count: arrays exceeding several thousand trapped atoms have been demonstrated, and Inflection's own roadmap targets 100 logical qubits by 2028 on its Sqale platform. Google's public framing of its own neutral-atom research program captures this precisely: superconducting processors are easier to scale in the time dimension (circuit depth), while neutral atoms are easier to scale in the space dimension (qubit count). This is the basis for the "Strong" C1 rating neutral atoms already carry in the §1.4 table, and Inflection's commercial momentum is consistent with it.

The second claim is throughput, and this is where the modality does not transfer to the present problem class. Neutral-atom gate operations require physical qubit shuttling - optical tweezers must move atoms into proximity before an entangling gate can execute - and published two-qubit gate times sit near 400 nanoseconds, roughly an order of magnitude slower than superconducting gate times, with end-to-end computation reported at one-hundredth to one-thousandth the speed of superconducting systems. SqDRIFT, as established in §1.4.2, is a throughput-bound algorithm: its time-to-solution scales with the product of per-circuit execution time and sample count, not with active-space size in isolation. A modality that scales favorably in qubit count but unfavorably in per-circuit gate speed is solving the wrong half of the SqDRIFT cost function for this dissertation's purposes.

This sharpens, rather than duplicates, the trapped-ion analysis of §1.4.2. Trapped-ion systems trade slow gates for exceptionally long coherence times, preserving a real advantage for deep-circuit, fault-tolerance-oriented work. Neutral atoms do not carry an equivalent compensating advantage for the throughput-bound regime: coherence times are good but not in the trapped-ion class, and no SqDRIFT-class benchmark - or comparable sample-based quantum diagonalization result - has been published on neutral-atom hardware at the active-space scale this dissertation targets, which is why C7 remains rated "Adequate (different alg.)" rather than "Strong." Inflection's software stack, Superstaq, is a cross-hardware compilation and optimization layer; nothing in its public description indicates a Part 11-aligned provenance or audit-trail framework comparable to the P1-P9 schema specified in §06.iii, leaving C5 and C6 correctly rated "Emerging" rather than revised upward.

Inflection's public listing is genuine evidence that the neutral-atom modality is commercially de-risking - it is not evidence that the modality is better matched to this dissertation's throughput-bound workload. The decision recorded in §1.4 stands: superconducting qubits

remain the best-supported anchor for a SqDRIFT-class pipeline at submission, and neutral atoms are correctly positioned in §1.4's table as a modality whose comparative advantage lies in a different axis of scaling than the one this dissertation's algorithm is bound by.

§ 02 Research Questions — Four Questions.

Can an orchestration layer that is bifurcation-aware (workflow-portable across IBM-native and Nvidia-substrate quantum stacks) and Part-11-governed (emitting nine-element provenance records, P1–P9, as first-class artifacts) unify quantum computing, HPC, and AI backends into a single reproducible workflow for non-druggable NSCLC tumor-suppressor mutation simulation?

Does the orchestration architecture correctly execute a VQE workflow end-to-end — from NGS input through active space selection, qubit encoding, Hamiltonian simulation, error mitigation, and AI rescoring — producing outputs consistent with quantum chemistry benchmarks on a simulator backend?

What design patterns and controls are necessary and sufficient to bring quantum-computed molecular binding data into compliance with FDA 21 CFR Part 11 electronic records requirements?

How should complex molecular structural information and quantum simulation results be presented to cross-disciplinary stakeholders — oncologists, pharma R&D, hospital IT — who lack quantum computing expertise?

§ 03 Literature Review & Research Gaps — What Exists, What's Missing.

3.1 · The Classical → Quantum Lineage and the Tractability Ceiling

The theoretical case for simulating chemistry on quantum hardware forms an unbroken lineage. Feynman (1982) first argued that quantum systems are intractable to classical simulation and that a computer built of quantum elements could simulate them efficiently; Lloyd (1996) made this constructive, proving that a universal quantum computer can efficiently simulate any local quantum system through Trotterized time evolution. Abrams and Lloyd (1999) then supplied the phase-estimation algorithm that converts simulated dynamics into ground-state eigenvalues — the formal bridge from "simulate the system" to "extract its energy" — and Aspuru-Guzik et al. (2005) mapped that machinery onto real molecules. McArdle et al. (2020) review the modern near-term descendants of this lineage. This continuum, from physical principle to executable molecular algorithm, is the theoretical spine on which the present architecture rests.

The lineage matters because the classical alternative has a definite ceiling. Exact correlation methods — full configuration interaction and the CASSCF/CASCI family — scale combinatorially with active-space size (Helgaker, Jørgensen, & Olsen, 2000; Levine et al., 2021), placing a practical wall near roughly eighteen active electrons. Reiher et al. (2017) quantify the crossover concretely, showing that classically intractable electronic-structure problems become reachable on fault-tolerant quantum hardware. The C275F target of this dissertation (~88 qubits under Jordan–Wigner encoding) sits decisively beyond that classical wall — which is precisely why a quantum route is the appropriate one.

On the near-term algorithmic side, Peruzzo et al. (2014) introduced the variational quantum eigensolver (VQE) as the hybrid loop pairing a quantum expectation-value estimate with a classical optimizer; McClean et al. (2016) established its theoretical framework; and Preskill (2018) named the NISQ regime that makes such hybrid methods the leading near-term strategy. Kandala et al. (2017) demonstrated hardware-efficient VQE on a superconducting processor, Bauer et al. (2020) surveyed the broader resource landscape, and Tilly et al. (2022) consolidated VQE methods and best practices. The randomized-compilation branch — Campbell's (2019) qDRIFT and its stochastic-sampling refinements (Kiss et al., 2023) — supplies the stochastic state-sampling basis for the sample-based quantum diagonalization (SqDRIFT) step used in Phase 3B. None of this work addresses specific oncology mutation targets, clinical translation, or the systems infrastructure required to operationalize quantum chemistry results.

3.2 · TP53 Mutation Landscape

Joerger and Fersht (2016) provide the definitive structural analysis of TP53 mutant classes, distinguishing cavity-creating mutants (Y220C) from DNA-interface structural mutants (C275F) and direct contact mutants (R248W). No published work applies quantum simulation to TP53 DNA-interface structural mutants such as C275F. No computational pathway exists in the literature for C275F drug discovery.

Figure 3.1 documents the platform's input diversity, not its scientific scope. Each of the five PDB entries is a data source the orchestration architecture must consume: a machine-readable record whose coordinate geometry determines active-space size, qubit demand, and backend routing for that specific job. The five targets differ significantly across all three dimensions — from the small ankyrin-repeat scaffold of CDKN2A p16 (156 residues, PDB 2A5E) to the Kelch β -propeller of KEAP1 (302 residues, PDB 2FLU) to the zinc-coordinated DNA-binding domain of TP53 C275F (~88 qubits under Jordan–Wigner encoding). Managing that heterogeneity through a single governed pipeline — same provenance schema, same compliance output, same audit trail — without manual re-derivation per target is the information systems design problem this dissertation solves. The platform consumes these five records; it does not produce them.

3.3 · Quantum Computing in Drug Discovery

Cao et al. (2019) reviewed quantum computing applications in quantum chemistry. Santagati et al. (2024) demonstrated quantum simulation of small drug-like molecules on real hardware. No published work applies these methods to non-druggable NSCLC tumor-suppressor structural mutants or integrates them into a governed clinical workflow.

3.4 · Hybrid Orchestration Platforms

Qiskit Runtime (IBM, 2021) and Orquestra (Zapata Computing, 2020) represented the first wave of single-vendor quantum orchestration. The post-March-2026 landscape is materially different: Nvidia's CUDA-Q programming model and the NVQLink GPU ↔ QPU interconnect (announced October 2025, expanded April 14, 2026 with the Ising AI models) constitute a cross-vendor, cross-paradigm orchestration substrate, and hybrid drug-discovery pipelines on it have been demonstrated — IonQ with AstraZeneca, AWS, and Nvidia on Suzuki–Miyaura coupling (June 2025) and Quantinuum with Nvidia on ADAPT-GQE applied to imipramine. Both precedents target generic small-molecule chemistry, not oncology mutation simulation, and neither is designed against a regulatory-records framework. What remains absent is therefore not a generic cross-vendor orchestration platform — that gap has begun to close — but a cross-vendor orchestration architecture specifically targeting oncology tumor-suppressor mutation simulation, with Part-11-grade provenance machinery sufficient to bring quantum-derived binding data into a regulated drug-development workflow. This is the narrower, defensible gap that §06.ii and §06.iii jointly close.

The post-April-2026 landscape was further shaped on March 26, 2026 by Cleveland Clinic and IBM's published quantum-centric supercomputing (QCSC) workflow — the first protein-scale electronic-structure simulation under that paradigm, whose architectural vision of tightly coupled HPC and quantum resources was set out by Alexeev et al. (2024) — which decomposed the 303-atom Trp-cage miniprotein into clusters and routed the most entangled (6–33 MOs each) to IBM Heron r2 via Embedded Wave Function with Sample-based Quantum Diagonalization (SQD). The Cleveland Clinic precedent is a methodological neighbor to this dissertation's sqDRIFT anchor (§1.4.1) and a structural neighbor to the bifurcation-aware orchestration claim (§06.ii), but it targets generic biomolecular discovery, not oncology-specific non-druggable mutations, and operates without a clinical-regulatory governance layer (no Part 11, no P1–P9 provenance record, no ML-decoder lineage capture). The narrower gap closed by this dissertation — oncology-mutation-specific orchestration with workflow-embedded compliance machinery — is therefore unaffected by the Cleveland Clinic precedent; if anything, it is sharpened.

3.5 · Regulatory Frameworks for Quantum Data

FDA 21 CFR Part 11 governs electronic records in regulated industries. Existing guidance covers classical computational models but does not address quantum-computed outputs. The April 14, 2026 release of Nvidia's Ising — a 3D convolutional neural network operating as a learned pre-decoder that resolves common syndromes at runtime and escalates the remainder to a conventional global decoder — sharpens the regulatory gap rather than closing it: a regulated drug-discovery pipeline must now produce Part-11-grade records not only of probabilistic quantum sampling (statistical estimators over noisy hardware) but also of a learned probabilistic pre-decoder in the QEC loop (and of the global decoder it defers to), whose model identifier, weight checksum, training corpus, and inference-time configuration become first-class compliance artifacts. The pattern is not unprecedented — FDA's evolving Good Machine Learning Practice (GMLP) guidance already requires model provenance for Software-as-a-Medical-Device — but it has not been extended to the QEC layer of a quantum simulation pipeline. No published framework applies Part 11 controls to quantum-derived binding data, and none addresses ML-decoded quantum error correction in a regulated workflow. This is the regulatory gap that §06.iii closes; the eight-element provenance record (P1–P8) is extended with a ninth element (P9) to capture ML-decoder lineage explicitly.

3.6 · Enterprise Architecture and Scientific-Data Provenance

The discipline that supplies a vocabulary for designing and governing complex systems-of-systems is enterprise architecture, established by Zachman (1987) and operationalized today through the TOGAF Standard and its Architecture Development Method (The Open Group, 2022), with ArchiMate (The Open Group, 2022) as the companion modeling language and Lankhorst (2017) as the standard scholarly treatment. These frameworks have been applied to scientific computing — Pierantoni et al. (2023) derive a reference architecture for science gateways spanning clouds, grids, and HPC — but the literature stops at the classical substrate: no published work frames a hybrid classical → quantum scientific pipeline as a governed enterprise architecture (as of June 2026; see the novelty-verification note in §06). Conversely, the quantum-chemistry literature reviewed in §3.1 is essentially silent on architecture governance.

The governance layer that makes such an architecture auditable rests on mature provenance and reproducibility standards. The FAIR principles (Wilkinson et al., 2016) define the findability, accessibility, interoperability, and reusability criteria a records layer must satisfy; the W3C PROV ontology (W3C, 2013) supplies an interoperable data model for serializing a provenance chain; and the reproducibility programs of Sandve et al. (2013) and Stodden et al. (2016) specify the operational and policy disciplines for trustworthy computational results. These standards, however, were written for conventional computation. None has been extended to a probabilistic, sampling-based quantum pipeline operating under regulatory-grade controls (21 CFR Part 11). The P1–P9 provenance schema of this dissertation is the instrument that conjugates these two otherwise-disconnected literatures — enterprise-architecture governance and scientific-data provenance — onto the classical → quantum continuum.

§ 04 Theoretical Framework — The Intellectual Scaffolding.

Design Science Research (DSR)

Hevner et al. (2004). The primary research method: building an artifact and evaluating it against defined criteria. The artifact is the contribution — its completeness, coherence, novelty, and utility are the evaluation criteria, not the scale of its deployment.

BQP-Relevant Quantum Chemistry

Bernstein & Vazirani (1997); Abrams & Lloyd (1999); Aspuru-Guzik et al. (2005). The theoretical grounding for placing the fermionic Hamiltonian eigenvalue problem at C275F's active-space boundary within BQP under standard quantum-chemistry framing. Establishes why quantum hardware is the appropriate substrate — under current classical state-of-the-art — without claiming a formal classical-quantum separation that the complexity literature has not proven for this problem class.

Translational Informatics

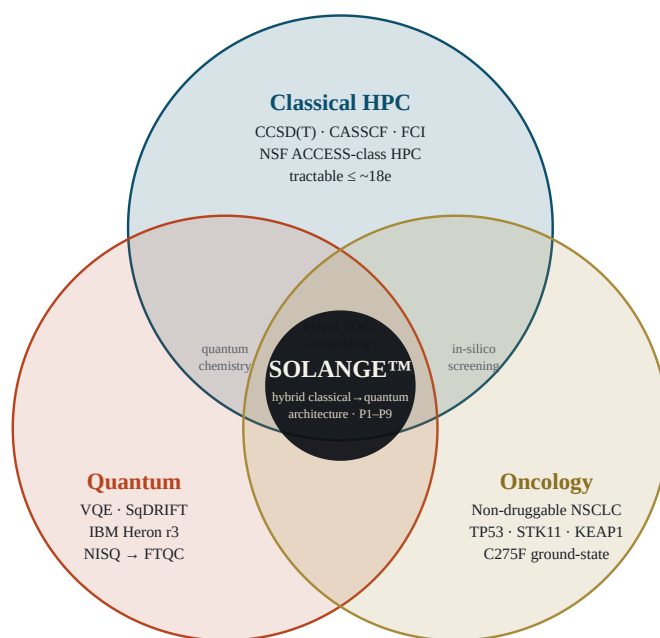
Sarkar (2010). The discipline of designing information systems that bridge clinical data, scientific computation, and actionable medical knowledge. Frames the platform as an IT artifact serving cross-disciplinary stakeholders rather than a physics instrument serving specialists.

Enterprise Architecture & the Classical–Quantum Continuum

The Open Group (2022); Lankhorst (2017). SOLANGE is not merely a quantum-chemistry pipeline but an enterprise architecture in the formal sense, and it is described here through TOGAF — the most widely adopted architecture framework. TOGAF contributes two instruments. First, a disciplined vocabulary that decomposes a complex socio-technical system into four architecture domains. Second, the Architecture Development Method (ADM), which models architecture not as a static blueprint but as a governed, iterative migration — the precise conceptual frame for SOLANGE's evolution from a classical proxy to quantum hardware.

TOGAF Domain	Framework Definition	SOLANGE Instantiation	Provenance
Business	Strategy, drivers, stakeholders, and the problem the system exists to solve.	Non-druggable NSCLC tumor-suppressor targets (TP53, STK11, KEAP1) — mutations with no small-molecule binding pocket.	—
Data	Logical and physical data assets and their governance.	Jordan–Wigner-encoded molecular Hamiltonians, CASSCF reference datasets, and the immutable P1–P9 provenance records (21 CFR §11.10(e)).	P1 · P2 · P5
Application	Applications, their behavior, and their interactions.	CASSCF/PySCF (classical reference), VQE on PennyLane (live Phase 3A), and SqDRIFT sample-based quantum diagonalization (Phase 3B).	P1 · P6
Technology	Hardware, networks, and the platform infrastructure beneath everything above.	The classical → quantum bridge itself: academic HPC (NSF ACCESS-class allocations) on one side, IBM Heron r3 on the other.	P3 · P4

The ADM frames Phase 3A → 3B as a single governed turn of the architecture cycle. A scientific subtlety is load-bearing here: the intermediate step targets a local binding site of roughly 24 active electrons (48 qubits) as its scientific ceiling — the precise scale at which both CCSD(T) classical chemistry (~18e) and full state-vector simulation (~4.5 PB) collapse. Operationally the run executes at ~20e/40q (~17.6 TB, within multi-node HPC reach on NSF ACCESS / USC CARC). In either case this is not classical chemistry: it is the variational quantum algorithm executed on a classical HPC simulator of the quantum circuit. The architectural objective — demonstrating that the governed pipeline scales to meaningful quantum-circuit width — is fully achieved at 40 qubits (see §8, Table 8.1 for the memory-scaling rationale). The provenance layer (P1–P9) operates throughout as TOGAF's Architecture Governance, rendering each iteration auditable. The contribution of this dissertation is located precisely at the intersection of three otherwise-mature fields — classical HPC, quantum computation, and computational oncology — where no prior system has operated under a unified, provenance-complete architecture.



[Figure — rendered from source]

§ 05 Research Methodology — Five Phases.

The methodology is structured as five phases of Design Science Research (Hevner et al., 2004) — building an artifact, evaluating it against defined criteria, and refining through expert review. The Phase 3 proof-of-concept anchors on superconducting hardware for the empirical-precedent reasons defended in §1.4; the architecture itself remains hardware-agnostic at the workflow level, with vendor-specific compilation and error-mitigation layers documented explicitly (§1.4).

Foundation

Patient NGS analysis · Literature verification · BQP-relevant classification argument · Active space determination (~44e, ~88 qubits)

Platform Prototype

Four-screen SOLANGE orchestration platform · Hardware-agnostic backend abstraction · FDA 21 CFR Part 11 compliance module · 3D molecular visualization · Deployed at solange-platform.bio

Architecture Validation Run · Minimal Active Space (4-qubit, live) → Local Binding Site (~20e/40q, pending HPC)

The objective of Phase 3A is to demonstrate that the orchestration layer executes end-to-end, not to produce a publishable binding-energy result. Minimal (live): 4-qubit VQE on PennyLane default.qubit, CAS(2e,2o)/STO-3G, AllSinglesDoubles UCCSD ansatz, 80 Adam steps — converges to within 1×10^{-5} Ha of CASSCF-exact in ~7 s on Render cloud. Every workflow stage (active-space selection, qubit encoding, Hamiltonian simulation, P1–P9 audit emission, SHA-256 P8 seal, Supabase persistence) is verified end-to-end. Intermediate (pending HPC): the same orchestration and governance pipeline is exercised at a larger active space of C275F on NSF ACCESS / USC CARC. The scientific ceiling target is ~24e/48q — precisely the scale at which both CCSD(T) classical chemistry (~18e) and full state-vector simulation (~4.5 PB) collapse — but the operational run executes at ~20e/40q via an exact state-vector simulation of the quantum circuit (~17.6 TB; not a tensor-network approximation), within multi-node HPC reach and already two orders of magnitude beyond the minimal prototype.

This intermediate run is a maturation of the orchestration and governance layer across increasing scale, not an extension of VQE toward hardware. The quantum algorithm is deliberately chosen by scale and substrate — VQE for the small, classically verifiable cases (Phase 3A), and SqDRIFT for the hardware-scale problem (Phase 3B) — mirroring the field's own transition; the architecture is designed to be agnostic to this swap, and the discontinuity between the two algorithms is a considered design choice rather than a claimed continuum. Two consequences follow. First, operational stability and scientific validity are separate axes. The Phase 3A claim is the former: that the governed pipeline (authentication, Hamiltonian encoding, quantum-kernel execution, P1–P9 provenance, SHA-256 sealing) runs end-to-end and stably at meaningful quantum-circuit width — an engineering result, fully met at 40 qubits. It is not a claim that the computed energy is scientifically correct: a run can execute stably and converge smoothly yet still converge to a wrong value if the active space or reference is flawed. Second, the 20-active-electron convergence is measured against an approximate classical reference. At the minimal 4-qubit CAS(2e,2o) scale the VQE energy converges to within 1×10^{-5} Ha of the CASSCF-exact value; at ~20e/40q there is no exact classical reference (full-CI is intractable at this active-space size), so convergence is assessed against an approximate reference (DMRG or CCSD(T)) — an approximate-against-approximate comparison, stated honestly as such. See §8 for the memory-scaling rationale.

Architecture Scaling Run · Full Binding Site (~44e)

The objective of Phase 3B is to demonstrate that the architecture scales without re-derivation when the active-space boundary changes. Re-execute the same orchestration workflow on the full binding site, incorporating the Zn^{2+} coordination environment (C176,

H179, C238, C242) and the DNA-guanine interface. The IT-side test is whether the audit trail, telemetry, and compliance record retain integrity across the boundary change. Requires approximately 88 qubits under Jordan-Wigner encoding, placing it within IBM Heron r3 capacity. Designated as a near-term post-proof-of-concept target, contingent on IBM Quantum Network institutional access.

The Transitional Window: Situating SqDRIFT and Heron r3 Within IBM's Fault-Tolerant Roadmap

The selection of IBM Heron r3 as the quantum hardware substrate and SqDRIFT as the molecular simulation algorithm reflects a deliberate positioning within a well-defined and time-bounded technological window. IBM's publicly disclosed roadmap — supported by a \$10+ billion capital commitment and a \$1 billion CHIPS Act award — establishes a clear architectural trajectory away from noisy intermediate-scale quantum (NISQ) devices toward fault-tolerant systems based on quantum low-density parity check (qLDPC) error correction. The milestones on this trajectory are specific: the Kookaburra processor (2026) introduces modular quantum memory, Cockatoo (2027) entangles multi-chip configurations, and Starling (2029) delivers the first large-scale fault-tolerant system at 200 logical qubits executing 100 million quantum gates.

This roadmap does not diminish the contribution of the present work — it contextualizes and justifies it. Heron r3, as the current-generation superconducting processor, and SqDRIFT, as a leading NISQ-era sample-based quantum diagonalization protocol whose stochastic state sampling inherits qDRIFT's formally bounded error scaling, represent the state of the art during precisely the window in which clinically relevant quantum molecular simulation first becomes computationally tractable. IBM's own 2026 demonstration (IBM Research; arXiv:2603.08696) applied SqDRIFT on 100 qubits to a 32-electron active space of $C_{13}Cl_2$ — a benchmark that for the first time brings the active-space requirements of NSCLC tumor-suppressor mutations within reach of available hardware.

The significance of this window extends beyond the chemistry. The transition to fault-tolerant quantum computing will not resolve — and in fact will intensify — the information systems challenges that constitute the primary contribution of this dissertation. As IBM scales toward Starling (2029) and the Blue Jay quantum-centric supercomputer (2,000 logical qubits, 2033), targeting drug discovery as an explicit commercial vertical, the regulatory reconstruction problem becomes more acute, not less. Probabilistic, learned quantum outputs will be generated at greater scale and complexity, yet no established framework exists for governing their audit trails, ensuring reproducibility, or satisfying FDA 21 CFR Part 11 requirements in a clinical submission context. The pipeline architecture, P1–P9 provenance schema, and bifurcation-aware orchestration developed in this dissertation are designed precisely for this trajectory: they establish the IS governance substrate that fault-tolerant quantum drug discovery workflows will require, built and validated during the window in which the underlying quantum computation first becomes feasible.

The present work therefore occupies a strategically necessary position: it is neither premature nor transitional. It is foundational.

Triangulated DSR Evaluation

Four converging evidence sources (technical, comparative, expert, case-based), executed in parallel sub-phases 4A–4F over Jan–Apr 2027. Detailed below.

Writing & Defense

Dissertation drafting May–Sep 2027 · Pre-defense committee revisions Oct 2027 · Target defense Q4 2027

Phase 4 · Triangulated DSR Evaluation Architecture

The artifact is evaluated through four independent evidence sources, following the FEDS framework (Venable et al., 2016) for DSR evaluation strategy. Each layer addresses a different validity threat; together they triangulate to produce a defensible aggregate judgment that no single source could supply alone.

Layer 1 · Technical Evaluation (Artifact Performance — No Human Subjects)

Objective measurement of the artifact itself, executed via an automated test harness with pre-registered acceptance thresholds.

Criterion	Metric	Acceptance Threshold
End-to-end workflow execution	% of pipeline stages completing without manual intervention across 50 simulated runs	≥ 95%
Audit-trail integrity (P1–P9)	% of runs producing complete, cryptographically valid provenance records	100%
Cross-vendor reproducibility	Δ between PennyLane and Qiskit Aer simulator outputs (χ^2 test on outcome distributions)	$p > 0.05$
Compilation-layer translation overhead	SWAP-gate count delta between superconducting and trapped-ion compilation targets	Documented & bounded
Active-space scaling integrity	Audit-trail completeness across local (~24e) → full (~44e) boundary change	100%
Compliance record validation	% of P1–P9 records passing automated Part 11 §11.10 validation checks	100%
Time-to-result	Wall-clock from NGS input to validated output, simulator backend	Measured & reported
Failure-mode coverage	% of injected fault scenarios (network drop, calibration drift, sampling truncation) detected by audit layer	≥ 90%

Layer 2 · Comparative Evaluation (Benchmarking vs. Existing Alternatives)

Reference task — a VQE simulation of a small molecule (H_2 or LiH) — executed through this platform and through three baseline pathways: Qiskit Runtime alone, Orquestra (Zapata), and fully manual workflow. Comparison metrics: number of manual steps, setup time, and whether the output satisfies 21 CFR Part 11 audit requirements.

Capability	This Platform	Qiskit Runtime	Orquestra	CUDA-Q + NVQLink	Manual
Multi-vendor QC backend abstraction	✓	✗ (IBM only)	Partial	✓ (cross-vendor, excludes IBM)	✗
HPC integration (GROMACS / ORCA)	✓	✗	✗	✓	Manual
AI rescoring layer	✓	✗	✗	Partial (CUDA-Q-native AI primitives)	Manual
21 CFR Part 11 audit trail	✓ (P1–P9)	✗	✗	✗	✗
Stakeholder visualization layer	✓	✗	✗	✗	✗
Statistical-equivalence validation	✓	✗	✗	✗	✗
Bifurcation-aware (across IBM-native & Nvidia substrate)	✓	✗	✗	✗ (one side of the bifurcation)	✗

Layer 3 · Expert Panel Evaluation (Stakeholder Validity)

Panel of $n = 4$ – 6 experts, composition deliberately calibrated for methodological coverage:

- 1 quantum chemistry / quantum computing expert (academia) — scientific and algorithmic rigor
- 1 oncology / molecular pathology clinician — clinical relevance and target validity
- 1 pharma R&D / computational drug discovery (industry) — translational practicality
- 1 Information Systems / DSR methodologist — methodological-validity guard (Hevner G5)
- 1–2 additional reviewers (flex) — FDA / regulatory affairs or hospital IT architect / clinical informatics

Composition aligned with dissertation-committee guidance ($n \leq 6$) and mapped to the Triangulated DSR layers: each layer of evaluation has at least one independent reviewer, with the methodologist guarding rigor across all four.

Evaluation protocol follows Hevner et al. (2004) — the Seven Guidelines, mapped to concrete instruments:

Guideline	Evaluation Question	Method
G1 · Design as artifact	Does the artifact exist, documented and operational?	Live demo + code review
G2 · Problem relevance	Is the problem significant to stakeholders?	Likert survey + open-ended response
G3 · Design evaluation	Does the artifact meet utility / quality / efficacy?	Triangulated (Layers 1–3)
G4 · Research contributions	Are the contributions novel and defensible?	Peer review by domain experts
G5 · Research rigor	Is the methodology rigorous?	DSR methodologist review
G6 · Design as search process	Are rejected alternatives documented?	Design-log review
G7 · Communication	Is the artifact transferable to stakeholders?	Stakeholder usability sessions

Scoring: Likert 1–7 on each criterion. Acceptance threshold: mean ≥ 5.0 with no individual score < 4 . Inter-rater reliability: Krippendorff's α reported on agreement-targeted criteria; $\alpha < 0.67$ flags a criterion requiring refinement or signaling genuine disagreement.

Layer 4 · Case-Based Walkthrough (Peffer et al., 2007)

Two clinical scenarios are traced end-to-end through the platform, one real (the anchoring patient — MI25-0349 with TP53 C275F), one hypothetical (a KEAP1 loss-of-function case). For each scenario the documentation captures: every design decision invoked; every point at which the audit trail caught — or should have caught — an anomaly; every step requiring manual intervention; and the stakeholder output produced for each persona (oncologist view, pharma view, IT view). This narrative complements the numbers — the case-based evidence is what shows the system works in context, not only on the bench.

Phase 4 Sub-Phases & Schedule

Sub-phase	Activity	Duration
4A	Layer 1 — automated technical-evaluation harness	4 weeks
4B	Layer 2 — comparative benchmarking	3 weeks
4C	Expert-panel recruitment & briefing materials (parallel to 4A/4B)	3 weeks
4D	Panel sessions — asynchronous, two rounds (theoretical + hands-on)	4 weeks
4E	Case-based walkthrough documentation	2 weeks
4F	Triangulation analysis & evaluation report	3 weeks

Total ~14–16 weeks, fitting the Jan–Apr 2027 window with 4A/4B/4C running in parallel.

Pre-Registration of Evaluation Criteria

All Layer-1 acceptance thresholds, Layer-3 Likert criteria, and Layer-4 walkthrough scenarios are pre-registered as a frozen document on OSF (Open Science Framework) prior to the start of Phase 4. This neutralizes the post-hoc-fit critique — that thresholds were adjusted after results were observed — and aligns the dissertation with the FEDS evaluation framework (Venable, Pries-Heje & Baskerville, 2016).

External Validity Statement

The evaluation is run on one mutation instance (TP53 C275F) with one quantum modality anchor (superconducting). Generalization to other mutations or other modalities is therefore analytical generalization in Yin's sense (2014) — an inference about the scope of the artifact's design principles, not statistical extrapolation from a sample. The architecture's mutation-agnostic and (workflow-level) hardware-agnostic design supports analytical generalization to BQP-relevant fermionic targets and gate-model quantum backends; statistical generalization is explicitly outside the dissertation's scope.

§ 06 Original Contributions — Three Claims.

This chapter advances three original Information Systems contributions. The biochemical setting — TP53 C275F, non-druggable tumor-suppressor mutations, electronic structure of binding sites — provides the anchor problem class, but the contributions themselves are IS-research artifacts: a classification framework, an orchestration architecture, and a regulatory-compliance schema. The dissertation does not aim to solve the biochemical problem of drug discovery directly; it builds the systems-engineering scaffold within which such work becomes governed, reproducible, and reusable.

Novelty-verification note. Each "first"/"no published work" claim in this chapter is stated as of June 2026. Priority was checked against Google Scholar, arXiv (quant-ph, cs.CE), PubMed, IEEE Xplore, and the Web of Science Core Collection, using combinations of the terms quantum · 21 CFR Part 11 / provenance / audit trail · drug discovery / oncology / tumor suppressor · orchestration / QC-AI-HPC, with a literature cutoff of June 2026. No prior work combining all elements of a given claim was found at that date; each claim is bounded to that evidence base and is revisable as the literature advances.

BQP-Relevant Classification Framework for Non-druggable Tumor Suppressor Targets

The first published analysis (as of June 2026) establishing TP53 C275F as a BQP-relevant problem in the context of oncology drug discovery — under the standard complexity-theoretic framing of quantum chemistry (Abrams & Lloyd, 1999; Aspuru-Guzik et al., 2005). The framework defines the active-space boundary (~44 electrons) beyond which the best known classical methods do not achieve the required chemical accuracy within tractable resources, and at which a known BQP-class quantum algorithm provides polynomial-time tractability. The framework is dual-conditional: classical intractability in practice plus BQP-class quantum tractability in principle. It does not claim BQP-hardness or a formal classical-vs-quantum separation, both of which remain open in general; it claims a defensible operational regime under which quantum-derived molecular evidence is the appropriate target of an orchestration architecture. The framework is explicitly conditional on the current state of classical methods (DMRG, neural quantum states, tensor networks) and revisable as those methods advance — providing a reusable classification template applicable to other non-druggable mutation targets. Operationally, the template sorts targets into three classes — the same classification the SOLANGE system applies in its live Non-Druggable Target Registry, where every mutation ingested from an NGS report is mapped and prioritized on arrival: Class A (quantum-necessary) — no classical path to chemical accuracy exists at the active-space boundary, so quantum simulation is the only route; Class B (quantum-advantaged) — classical methods are insufficient at the required accuracy, and quantum simulation supplies the advantage; and Class C (filtered out) — classical methods are sufficient, so the target is filtered out of the quantum pipeline as not quantum-relevant. The classification is thus a routing and prioritization policy — Class A and B enter the quantum queue; Class C is filtered out. Under this template C275F is Class B: it sits in the empirically quantum-advantaged regime defined by the current state of both classical and quantum methods circa 2026. This is a theoretical IT contribution; it does not require hardware execution to be valid.

The framework is designed as a reusable classification tool applicable beyond the four anchor mutations. Fifteen pipeline-eligible non-druggable mutations have been identified across eleven genes (TP53, KEAP1, CDKN2A, STK11, NF1, RB1, SMARCA4, RBM10, ARID1A, SETD2, U2AF1), collectively affecting an estimated 1.1–1.3 million unique NSCLC patients annually — representing the largest unaddressed population in computational oncology within the quantum feasibility window. Each satisfies the dual BQP-relevant condition: classical intractability at chemical accuracy at the specified active-space boundary, and a known BQP-class quantum algorithmic formulation for the corresponding fermionic Hamiltonian eigenvalue problem. The classification template defined here for C275F is directly applicable to all fifteen.

Part-11-Governed QC·AI·HPC Orchestration for Oncology Mutation Simulation

A cross-vendor orchestration architecture designed specifically for oncology tumor-suppressor mutation simulation, in which 21 CFR Part 11 provenance is emitted at workflow execution time rather than reconstructed retrospectively. As established in §3.4, the post-April-2026 landscape contains a generic cross-vendor substrate (CUDA-Q + NVQLink) that this architecture sits on top of where IBM's path permits, and across where it does not — Qiskit Runtime when running on IBM Heron, CUDA-Q-equivalent paths if a non-IBM modality is later substituted (§1.4.1). The contribution is not the cross-vendor abstraction itself, which is now commodity, but three properties the commodity substrate does not provide:

(1) Oncology-mutation-specific construction. The pipeline is built around NGS-driven active-space selection, mutation-aware Hamiltonian assembly, and AI rescoring of binding outputs for tumor-suppressor targets — not generic small-molecule chemistry of the kind exercised by IonQ/AstraZeneca on Suzuki–Miyaura coupling or Quantinuum/Nvidia on imipramine.

(2) Workflow-embedded Part-11 governance. The eight-element provenance record specified in §06.iii (extended to a ninth element P9 covering ML-decoded QEC, per §3.5) is emitted as a first-class artifact of workflow execution — capturing calibration epoch, per-gate error budget, and raw outcome distribution at the moment they are recoverable. Adding such records retrospectively is non-physical for probabilistic quantum simulation; the governance must be in the orchestration, not bolted onto its outputs.

(3) Selective paradigm invocation, bifurcation-aware. The workflow abstraction encodes a decision rule for when subproblems are dispatched to AI, HPC, or QC backends — based on problem characteristics such as active-space size, classical-method tractability under §06.i's BQP-relevance classification, and ML-decoder availability. The same abstraction does not commit the dissertation to a single vendor stack as the IBM Quantum-Centric Supercomputing path and the Nvidia NVQLink/CUDA-Q substrate continue to diverge; re-anchoring on a different modality, should the empirical landscape shift, requires re-implementing only the compilation and error-mitigation layer — not the workflow. The orchestration is therefore not merely cross-paradigm execution but selective computational escalation : AI handles pattern-detection and rescoring; HPC handles coarse classical simulation and conformational filtering; QC is invoked only where classical methods exceed practical limits at the active-space boundary.

The orchestration remains hardware-agnostic at the workflow level and vendor-specific at the compilation and error-mitigation layer , documenting — rather than concealing — the translation overhead these layers require. The first cross-vendor, Part-11-governed orchestration architecture (as of June 2026) designed specifically for oncology mutation targets. Validated by proof-of-concept execution on PennyLane simulator. Designed for IBM Heron r3 execution in the subsequent research phase.

A further consequence of this architecture is frequently underweighted in comparisons against precedent systems: the absence of inter-tool hand-offs is itself a governance property, not merely an operational convenience. The workflow described above — NGS-driven active-space selection, mutation-aware Hamiltonian assembly, classical reference computation, quantum circuit execution, error mitigation, and AI rescoring — would, absent a unifying orchestration layer, require a researcher to coordinate five or six independently maintained tools (a classical electronic-structure package, a quantum SDK specific to the target vendor, a separate quantum-sampling toolchain for Phase 3B, an HPC scheduler, and an ad hoc record-keeping system reconstructed after the fact). Each hand-off between such tools is a point at which the provenance chain specified in §06.iii can break silently: a result re-entered by hand, a parameter copied without its calibration context, a run whose device state was never captured because the capturing tool did not exist at that boundary. The single-platform design is therefore not offered as a usability feature distinct from the Part-11 claim of §06.iii — it is the precondition for that claim. P1–P9 records can be emitted as first-class artifacts of workflow execution (§06.ii) only because no boundary exists at which provenance capture is optional or manual. None of the precedent systems reviewed in §3.4 — including the architecturally proximate Cleveland Clinic/IBM QCSC workflow —

operate as a single governed platform across this full pipeline; each composes independently maintained tools without a provenance-preserving substrate spanning the composition.

FDA 21 CFR Part 11 Compliance Framework for Probabilistic Quantum Simulation

21 CFR Part 11 was drafted under the assumption that computational outputs are deterministic and reproducible bit-for-bit. Quantum simulation breaks that assumption at a foundational level: identical input circuits, executed on the same physical device under nominally identical conditions, yield distributions of measurement outcomes, and those distributions themselves shift with hardware noise, decoherence, and calibration drift. The April 2026 emergence of ML-decoded quantum error correction (Nvidia Ising) introduces a second-order break: the decoder itself is now a learned, probabilistic model whose lineage must enter the compliance record (§3.5). This contribution closes the regulatory gap with a defensible specification of what records must be retained, in what form, for quantum simulation outputs to satisfy Part 11.

Five Sources of Output Variability

The framework enumerates the irreducible sources of variance in a quantum simulation output, each with distinct provenance characteristics: (i) statistical sampling variance ($1/\sqrt{N}$ convergence of estimators over shot count — a property to be documented, not eliminated); (ii) hardware noise (gate infidelity, readout error, crosstalk — captured at execution time, not retrospectively); (iii) decoherence (T_1/T_2 recorded as device-state metadata at the calibration epoch); (iv) calibration drift (the boundary condition that defines, in operational terms, which physical device executed the simulation); (v) ML-decoder inference variance (when QEC is performed by a learned model rather than a deterministic algorithm — Ising-class decoders introduce inference-time precision effects, batch-ordering sensitivity, and version drift across model retrainings, each requiring distinct provenance treatment).

Nine-Element Provenance Record (P1–P9)

Compliance is achieved through a nine-element record attached to every simulation output that may be cited in regulatory submissions. P1–P8 are required for any quantum simulation execution; P9 is invoked whenever the QEC layer involves a learned decoder (e.g., a Nvidia Ising-class pre-decoder or successor models), in which case it records both the learned pre-decoder and the conventional global decoder it escalates to, and degrades to a minimal record (algorithm name + software version) for purely deterministic decoding (pyMatching, BP-OSD). Phase 3 of this dissertation runs on Qiskit with deterministic decoding and does not execute ML-decoded QEC at all; P9 is therefore a forward-looking design contribution — a specification of the records a regulated workflow will require once learned decoders enter production — not a component exercised by the present proof-of-concept.

P1	Circuit fingerprint — cryptographic hash of the post-transpilation compiled circuit
P2	Compilation lineage — source Hamiltonian, fermion-to-qubit mapping, Trotter order/step, SqDRIFT sampling parameters
P3	Device identifier & calibration epoch — backend, processor revision, calibration timestamp
P4	Per-gate error budget snapshot — 1- and 2-qubit fidelities, readout-error matrices, T_1/T_2 at the calibration epoch
P5	Shot count & raw outcome distribution — full bitstring-level histogram, permitting independent re-derivation of any aggregated estimator
P6	Error-mitigation lineage — every transformation applied to raw outcomes (ZNE, PEC, twirling), with parameters and method versions
P7	Statistical estimator & confidence interval — final reported observable, estimator function, 95% CI bounded by shot-count and noise-model variance
P8	Cryptographic seal — time-stamped digital signature over P1–P7 and P9, anchoring the record against subsequent modification
P9	ML-decoder lineage — when QEC inference uses a learned pre-decoder: decoder identifier & version, weights checksum, training-corpus manifest hash, inference-time configuration (precision, batch size, decoder threshold), execution environment (GPU class, CUDA/runtime version), and the conventional global decoder it escalates to (algorithm + version). For purely deterministic decoding: algorithm name and software version. Forward-looking — not exercised in Phase 3.

Why This Differs from Classical Monte Carlo

A natural objection is that classical computation already produces probabilistic outputs: Monte Carlo simulations are stochastic, and Part 11 already accommodates them through pseudo-random-number-generator seed logging — record the seed and the run is reproducible bit-for-bit. What, then, is new here beyond documenting an RNG? The answer is that quantum stochasticity is not seed-controllable. A classical Monte Carlo trajectory is fixed by a single logged integer, so re-execution reproduces it exactly on demand. A quantum run is not: hardware noise, decoherence, and calibration drift are physical processes of the device at execution time, not a controllable parameter, and no logged value can reproduce an identical run even in principle — re-executing the same circuit on the same processor an hour later samples a different noise realization, and after a recalibration it is, for provenance purposes, a different device. This is precisely why §11.10(a)'s bit-for-bit reproducibility test is broken by quantum simulation in a way it is not broken by Monte Carlo, and why the statistical-equivalence reinterpretation below is a genuine extension of Part 11 rather than a restatement of existing seed-logging practice. The ML-decoder layer (P9) is a hybrid case: the learned decoder's weights and version are loggable and reproducible, but its inputs — live syndrome streams from a noisy device — are not; P9 therefore records the reproducible half (model lineage) alongside the irreproducible half (the device state that produced its inputs).

The Critical Reinterpretation

The framework's most consequential move concerns the meaning of "validation" under §11.10(a). For deterministic systems, validation tests verify that re-execution produces identical outputs. For probabilistic quantum simulation this test is non-physical and would mark every legitimate run as anomalous. The framework redefines validation as statistical equivalence: a re-execution is consistent with the original record if and only if the new outcome distribution lies within the confidence interval declared in P7, conditional on the same calibration epoch (P3), circuit fingerprint (P1), and decoder lineage (P9). Where any of these differs — calibration epoch, circuit, or decoder — the comparison becomes a hypothesis test on distribution overlap accounting for declared inter-condition variance drift. This is not a relaxation of the regulation; it is the only operationalization of §11.10(a) consistent with quantum mechanics — preserving Part 11's core intent (detecting unauthorized modification, system failure, or invalid records) while specifying the statistical machinery that must replace bitwise comparison.

§ 07 Preliminary Work — A Deployed Prototype.

A fully functional prototype platform is deployed and accessible. All research artifacts are version-controlled on GitHub and directly accessible below.

7.1 · Deployed Artifacts & Research Outputs

- SOLANGE Platform - Live prototype + 3D simulation
- SOLANGE Platform Architecture
- SOLANGE Operational Stack

The prototype demonstrates the complete pipeline flow, grounded in the actual patient NGS report MI25-0349. Simulation results are scientifically informed illustrative values. The 3D visualization is a schematic representation of known TP53 secondary structure, loop topology, and zinc coordination — not crystallographic data. Both are sufficient to demonstrate the architecture's design logic and stakeholder communication capability.

§ 08 Limitations & Scope — What This Is, And Isn't.

Simulation Results

Binding energy values in the current prototype are scientifically informed estimates, not outputs of executed quantum simulations. Phase 3 proof-of-concept will produce real computed outputs on a simulator backend.

3D Visualization

Atomic coordinates are procedurally generated and schematic; they correctly represent structural topology but are not derived from PDB crystallographic data.

Hardware Scale & Classical Simulability

Full 44-electron pocket simulation on IBM Heron r3 is future work. The dissertation contribution is the architecture that enables this execution, not the execution itself. Full-scale quantum simulation of TP53 C275F (~44 electrons, ~88 qubits) on IBM Heron r3 is designated as a post-dissertation research phase, contingent on IBM Quantum Network institutional access. The architecture is hardware-agnostic at the workflow level and vendor-specific at the compilation and error-mitigation layer — IBM Heron r3 is one supported backend among several, with documented translation overhead at the lower layers.

The table below quantifies why classical simulation of quantum circuits becomes intractable as qubit count grows, and contextualises the Phase 3A scaling choices. A quantum circuit on n qubits requires a complex state-vector of length 2^n ; stored in double precision (complex128, 16 bytes per amplitude), the memory footprint is $2^n \times 16$ bytes. Every additional qubit doubles the requirement.

Qubits n	State-vector size	Memory (complex128)	Hardware regime / Phase 3 role
4	16 amplitudes	256 B	Laptop / cloud VM · Phase 3A minimal — live
30	$\sim 1 \times 10^9$	16 GB	Single GPU / workstation
40	$\sim 1.1 \times 10^{12}$	~ 17.6 TB	Multi-node HPC (NSF ACCESS / USC CARC) · Phase 3A intermediate — operational target (~20e/40q)
45	$\sim 35 \times 10^{12}$	~ 563 TB	Leadership-class HPC (Frontier, Fugaku)
48	$\sim 281 \times 10^{12}$	~ 4.5 PB	Edge of global classical capacity · Phase 3A intermediate — scientific ceiling (~24e/48q)
88	$\sim 3 \times 10^{26}$	$\sim 4.6 \times 10^9$ PB	Physically intractable on any classical machine · Phase 3B — requires IBM Heron r3 QPU (~44e full site)

Phase 3A intermediate targets ~24e/48q as its scientific scale — the precise point where both classical chemistry (CCSD(T) collapses past ~18e) and classical state-vector simulation (~4.5 PB) reach their respective limits. For operational execution, the active space is reduced to ~20e/40q (~17.6 TB) via an exact state-vector simulation of the quantum circuit (not a tensor-network approximation), which is within multi-node HPC reach on NSF ACCESS or USC CARC. This reduction does not compromise the architectural objective, which is operational stability rather than scientific validity: the pipeline is demonstrated at quantum-circuit scale two orders of magnitude beyond the minimal prototype, under full P1–P9 provenance governance. Phase 3B (~44e/88q on IBM Heron r3) exits the classically simulable regime entirely — the point at which the quantum algorithm is deliberately swapped from VQE to SqDRIFT, and the reason quantum hardware is required.

Regulatory Scope

The compliance framework addresses data governance and audit trail requirements. It does not constitute an FDA submission or clinical trial regulation.

Generalizability

The platform is designed around NSCLC mutations but the orchestration architecture is mutation-agnostic and applicable to any BQP-relevant molecular simulation target.

Single-Modality Anchor

The proof-of-concept execution anchors on superconducting qubits (§1.4). This decision is the best-supported choice at the time of submission, but it narrows the empirical claims of the dissertation. Multi-modality validation — re-running the pipeline against trapped-ion or neutral-atom backends to demonstrate behavioral equivalence — is deferred to future work. The architecture is described abstractly enough that re-anchoring on a different modality is a defined exercise should the empirical landscape shift before defense.

Nvidia Substrate (Forward-Compatible, Not Empirically Exercised)

The post-April-2026 hybrid substrate — Nvidia CUDA-Q , NVQLink , and Ising — is acknowledged in §1.4.1 and treated as commodity infrastructure in the three-layer architecture (§06.ii). Phase 3 of this dissertation does not execute on that substrate. IBM is absent from Nvidia's announced NVQLink partner roster (Atom Computing, IonQ, IQM, Q-CTRL, Academia Sinica, EeroQ), and the SqDRIFT empirical anchor ($C_{13}Cl_2$, IBM Research 2026; arXiv:2603.08696) runs on IBM Heron via Qiskit Runtime with deterministic decoding (pyMatching). The contribution is therefore structural , not empirical: (a) the orchestration architecture is designed to interoperate with the Nvidia substrate (§06.ii property 3, bifurcation-aware), but Phase 3 exercises only the IBM-native side; (b) the ninth element P9 of the Part-11 provenance record (§06.iii) is forward-compatible — it specifies what records a regulated workflow will require when ML-decoded QEC enters production, but Phase 3's P9 records reduce to the minimal deterministic form (algorithm name + software version). Empirical validation of bifurcation-aware orchestration across both stacks, and of ML-decoder provenance on a live Ising-class decoder, are designated post-dissertation research phases.

IND Regulatory Boundary

The compliance framework (§06.iii) addresses the information-systems layer of FDA 21 CFR Part 11 — what records must be retained, in what form, against what validation criteria. It does not address the broader regulatory-science questions: the criteria under which quantum-simulated binding affinities could be admitted as evidence in an Investigational New Drug (IND) submission, clinical-trial protocol design implications of using quantum-derived candidates, or FDA's separate guidances on Model-Informed Drug Development. These topics fall outside the dissertation's scope and are reserved for future regulatory-science work.

§ 09 Significance — Why This Matters.

Of the 1.53 million patients who die annually from NSCLC, a significant population carries non-druggable mutations for whom no computational drug discovery pathway currently exists. Quantum hardware has reached sufficient capacity. The algorithms exist. The clinical data exists. What is missing is the information systems infrastructure that connects them — governed, reproducible, and accessible to the cross-disciplinary stakeholders who need to act on the results. The platform is designed around four NSCLC anchor mutations but the architecture is extensible to over one million patients annually across fifteen pipeline-eligible non-druggable mutations — the largest unaddressed population in computational oncology within the quantum feasibility window.

This dissertation builds that infrastructure. Not as a physics paper claiming a new binding energy, but as an IT systems contribution claiming a new architecture. The distinction is precise and important:

Translational Pipeline — Where the Architecture Sits in the Larger Pathway

This subsection situates the IT contribution inside the wider drug-development arc; the downstream stages (in vitro through Phase III) are not claimed as dissertation contributions. The architecture's role is bounded to the In Silico stage — providing governed, reproducible, audit-traceable computational outputs that downstream stages can cite as a primary record. The clinical and regulatory stages that follow are shown to make the IT contribution's significance concrete, not to extend the dissertation's scope into pharmacology or clinical trials.

This Dissertation

Quantum simulation of C275F local (~24e) and full (~44e) binding sites using the QC·AI·HPC orchestration platform. Output: ranked drug candidate molecules with quantum-accurate binding energies, governed by FDA 21 CFR Part 11-compliant audit trails.

Biochemical Validation

p53 binding assays, transcriptional activation assays, and cell-free DNA binding experiments on shortlisted candidates. Confirms that computationally predicted binding translates to measurable biochemical activity.

Animal Model Studies

Efficacy and preliminary safety profiling in murine lung cancer models harboring the C275F mutation. Provides pharmacokinetic and toxicity data required before human studies.

Human Safety & Dosing

First-in-human studies establishing safety profile, maximum tolerated dose, and pharmacokinetics in cancer patients.

Efficacy Signals

Targeted enrollment of TP53 C275F-positive NSCLC patients. Primary endpoint: objective response rate. First clinical evidence that quantum-designed candidates produce measurable anti-tumor activity.

Regulatory Approval

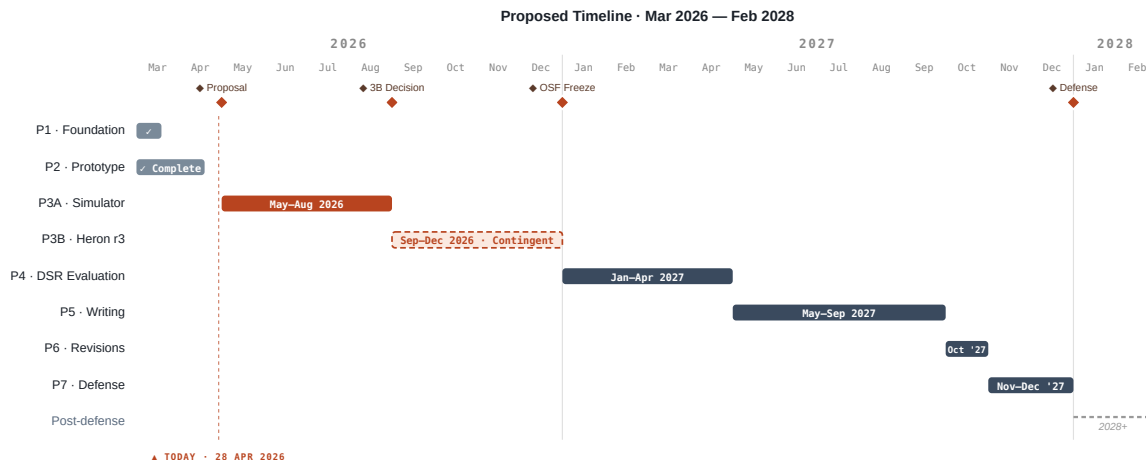
Randomized controlled trials against standard of care. Successful completion enables FDA submission and clinical deployment — completing the translational arc from quantum simulation to approved therapy for a currently non-druggable mutation.

This framing positions the quantum simulation architecture not as an isolated computational exercise, but as the critical first step in a clinically meaningful and scientifically grounded drug discovery pipeline targeting currently non-druggable TP53 mutations in lung cancer.

Future Research Directions

The molecular-level simulations enabled by this dissertation's SqDRIFT+VQE pipeline are positioned to inform a second clinical question that lies beyond the dissertation's scope: NGS-informed radiotherapy decision support . The non-druggable mutations targeted here — TP53, KEAP1, CDKN2A, STK11 — are the same mutations most frequently implicated in radioresistance literature, and the quantum-derived mechanistic features they expose (DNA damage repair dynamics, oxidative-stress response via the NRF2 pathway disrupted by KEAP1 mutations, apoptotic signaling) are precisely the features current radiogenomic models lack. Bridging quantum-simulated mutation effects into radiotherapy planning — extending prior efforts such as the Genomic-Adjusted Radiation Dose (GARD) framework (Scott et al., 2017) — is a defensible post-doctoral research program that this dissertation enables but does not undertake. The dissertation builds the molecular-simulation infrastructure; the radiotherapy decision-support layer, with its independent regulatory pathway and prospective-validation requirements, is reserved for future work.

§ 10 Proposed Timeline — The Road Ahead.



[Figure — rendered from source]

Phase	Activity	Status	Target
1	Foundation and literature review	✓ Complete	—
2	Platform prototype — deployed	✓ Complete	—
3A	Architecture validation on PennyLane simulator — minimal: 4-qubit CAS(2e,2o) ✓ live; intermediate: local site scientific target ~24e/48q (classical-chemistry ceiling + state-vector memory limit ~4.5 PB); operational run ~20e/40q (~17 TB, within HPC reach) ► pending	✓ Partial	May–Aug 2026
3B	Architecture scaling run — full site (~44e) on IBM Heron r3 (contingent on IBM Quantum Network access)	Contingent	Sep–Dec 2026
4	Triangulated DSR evaluation — Layers 1–4 across sub-phases 4A–4F (technical, comparative, expert panel n=4–6, case-based walkthrough)	Pending	Jan–Apr 2027
5	Dissertation writing (chapters 1–8)	Pending	May–Sep 2027
6	Pre-defense committee revisions	Pending	Oct 2027
7	Defense	Pending	Nov–Dec 2027
—	IBM Heron r3 full-scale runs (post-defense)	Future work	2028+

Timeline Rationale & Contingencies

The timeline allocates ~20 months from proposal acceptance to defense — a deliberate revision from earlier drafts that compressed the schedule into 9 months. The realistic distribution recognizes that (a) Phase 3A is a four-month engineering effort, not a two-week sprint; (b) Phase 3B is contingent on IBM Quantum Network institutional access, which can take 3–6 months to secure; (c) recruiting 4–6 cross-disciplinary domain experts and running structured DSR evaluation cannot be compressed below one quarter without losing methodological rigor; and (d) writing a defensible IT/DSR dissertation while balancing professional commitments requires a sustained five-month drafting window plus a one-month revision cycle.

Phase 3B is explicitly designated as contingent. The dissertation's central contribution — the orchestration architecture and the FDA Part 11 reinterpretation — is independent of full-pocket execution on physical quantum hardware. If IBM Quantum Network access is not granted within the Sep–Dec 2026 window, Phase 3B is documented as future work in the limitations chapter, and the defense proceeds on

the strength of Phase 3A (architecture validation) and Phase 4 (expert panel evaluation). This contingency is the practical expression of the "architecture is the contribution; simulation is the application" framing.

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Appendix A Complete Glossary — Seventy-Seven Entries.

A Translational QC·AI·HPC Orchestration Platform for Non-druggable NSCLC Tumor-Suppressor Mutations

This glossary consolidates every acronym and technical term that appears across the dissertation proposal, conference poster, deployed prototype (solange-platform.bio), and supporting coursework artifacts. Each entry pairs the standard definition with the term's specific role in this dissertation, so a reader unfamiliar with quantum computing, computational chemistry, FDA regulation, or oncology can follow the work without external reference. Terms are listed alphabetically.

Total entries: 77

Acronym	Full Name	Definition & Context in This Dissertation
21 CFR	Title 21, Code of Federal Regulations	The U.S. federal regulations governing food and drugs, administered by the FDA. Two parts are relevant: Part 11 (electronic records - in dissertation scope) and Part 312 (IND submission - explicitly out of scope).
3D CNN	Three-Dimensional Convolutional Neural Network	A CNN extended to three-dimensional input grids. Architecture of the Nvidia Ising QEC decoder, applied to syndrome lattices in real-time error correction. Layer 2 commodity; provenance recorded in P9.
ALK	Anaplastic Lymphoma Kinase	An oncogene whose rearrangements drive ~5% of NSCLC cases. ALK has FDA-approved targeted therapies (crizotinib, alectinib) and therefore represents the druggable counterpart to the tumor-suppressor mutations addressed in this dissertation. Cited in the Problem Statement to delineate the clinical gap.
API	Application Programming Interface	A defined contract through which software components communicate. Referenced in problem statement P1 (no standardized handoff between classical and quantum compute) - the absence of a stable API contract between HPC pre-screening and quantum circuit builders is one of the gaps the orchestration platform addresses.
ATP	Adenosine Triphosphate	The principal energy-carrying molecule in cells; binds to kinase active sites. Relevant to STK11 - one of the four target mutations - whose kinase domain coordinates ATP and contributes to the active-space electron count requiring quantum-grade simulation.
B3LYP	Becke 3-parameter Lee-Yang-Parr (DFT functional)	A widely used hybrid density functional in classical computational chemistry. The poster's results section reports B3LYP at -8.4 kcal/mol with 2.8 kcal/mol error - outside the chemical accuracy threshold of ≤ 2 kcal/mol - establishing why classical DFT is insufficient for C275F at the active-space boundary.
BAA	Business Associate Agreement	A HIPAA-mandated contract between a covered entity (hospital, clinic) and any external party that handles Protected Health Information on its behalf. Out of scope for this dissertation because the platform uses fully de-identified single-case reference data (MI25-0349) but would become relevant in any future multi-patient extension.
BPR	Business Process Reengineering	A management methodology for analyzing and redesigning organizational workflows. Used in the IST 697 coursework deliverable that mapped the QC+AI+HPC drug discovery platform across five business processes (P1–P5), identifying the Molecular Simulation Pipeline as the main process where the BQP-class computational bottleneck concentrates.
BQP	Bounded-error Quantum Polynomial time	The complexity class of decision problems efficiently solvable on a quantum computer with bounded error probability. The dissertation's first original contribution (§06.1) is a BQP-relevance classification framework arguing that the fermionic Hamiltonian eigenvalue problem at C275F's 44-electron active-space boundary admits a known BQP-class quantum algorithm while best-known classical methods fail at chemical accuracy. This is an operational BQP-relevance claim — conditioned on the current classical state of the art and on sufficient initial-state overlap — not a formal classical–quantum separation, which remains open (ground-state finding is QMA-hard in general; see §1.2).
C ₁₃ Cl ₂	13-Carbon, 2-Chlorine half-Möbius molecule	The synthetic molecule used by IBM in its 2026 demonstration (IBM Research; arXiv:2603.08696) to run SqDRIFT at 32 electrons / 100 qubits - the empirical anchor of the dissertation's Layer 1 scientific claim. The transition from C ₁₃ Cl ₂ (synthetic, vacuum, cryogenic) to a clinically relevant metalloprotein mutation is explicitly framed as Layer 1 of the dissertation's novelty argument.
C275F	TP53 p.Cys275Phe (cysteine-to-phenylalanine substitution at position 275)	A non-druggable TP53 missense mutation in which Cys275 is replaced by Phe. The mutation introduces a rigid aromatic π -system into the hydrophobic β -sandwich core of the p53 DNA-binding domain, disrupting DNA-contact geometry without creating an exploitable drug-binding cavity. C275F is the dissertation's primary demonstration case - sourced from patient NGS report MI25-0349 - and the focal point of the Phase 3A and 3B simulations. C275F is classified as a structural mutant: the Cys → Phe substitution destabilizes the hydrophobic core of the DNA-binding domain rather than directly abolishing a DNA-phosphate contact. No crystallographic structure of C275F exists in PDB, consistent with the conformational heterogeneity expected from a structural mutation.
CASSCF	Complete Active Space Self-Consistent Field	A multireference quantum chemistry method that treats a chosen subset of orbitals (the active space) with full configuration-interaction accuracy. Cited as one of the classical baselines: CASSCF becomes intractable beyond approximately 18 active electrons, which is exactly why the four target mutations (TP53, KEAP1, CDKN2A, STK11) at 14–28 electrons sit at or beyond the classical ceiling.
CCSD(T)	Coupled Cluster with Single, Double, and perturbative Triple excitations	The classical "gold standard" of quantum chemistry for single-reference systems, with O(N ⁷) computational scaling. The poster lists CCSD(T) as infeasible at 44 active electrons and uses it as the reference value in the benchmarking comparison (~11.2 kcal/mol). VQE-SqDRIFT outputs are validated against published CCSD(T) values where available.
CDKN2A	Cyclin-Dependent Kinase Inhibitor 2A	A tumor-suppressor gene encoding p16 and p14ARF. CDKN2A loss is one of the four target mutations in the dissertation. Its hydrophobic core disruption requires an active space in the 14–18 electron range, straddling the classical tractability threshold and providing a less metallic counterpoint to TP53 in the dissertation's mutation portfolio.
CMC	Chemistry, Manufacturing, and Controls	The section of an FDA Investigational New Drug (IND) submission documenting drug composition, synthesis, and quality control. Out of dissertation scope (Part 312 territory) but mentioned in context to clarify the boundary between this dissertation's Part 11 contribution and the broader regulatory pathway.
CNN	Convolutional Neural Network	A deep-learning architecture specialized for grid-structured data such as images. Nvidia Ising's QEC decoder is implemented as a 3D CNN that acts as a fast pre-decoder across a three-dimensional syndrome lattice, resolving common errors in real time and escalating the remainder to a conventional global decoder. Referenced in Layer 2 of the architecture; the provenance of both stages is captured in field P9 of the audit trail (a forward-looking field — Phase 3 uses deterministic decoding).
CRO	Contract Research Organization	An external company providing outsourced research services to pharma. Referenced in the BPR analysis (problem P3) - current CRO handoffs proceed without formal go/no-go thresholds even when confidence intervals are low, illustrating the governance gap that the orchestration platform addresses.
CSA	Computer Software Assurance	FDA guidance (finalized September 2025) requiring risk-based validation of software used in regulated contexts. The dissertation's Layer 5 framework maps each pipeline component (Qiskit compiler, SqDRIFT sampler, AI scoring module, uncertainty quantifier) to CSA-compliant validation documentation.
CUDA-Q	Nvidia's quantum-classical programming model	A programming framework enabling unified development across GPU and QPU resources. Listed in Layer 2 (commodity substrate) of the architecture. Its emergence as a vendor-spanning standard is what reframed the dissertation's claim from

		"cross-vendor abstraction" (now commodity) to "Part-11-governed, bifurcation-aware orchestration" (defensible white space).
DBD	DNA-Binding Domain	The structural region of p53 (residues ~94–312) responsible for sequence-specific DNA recognition. C275 lies within the DBD; its mutation to Phe disrupts the loop-sheet-helix DNA-contact interface. The DBD is the spatial anchor for active-space selection in the dissertation.
DeepChem	DeepChem (open-source ML library for chemistry)	A Python toolkit providing pre-built deep-learning models for drug discovery, including Graph Neural Networks. Used in Layer 2 for AI rescoring of quantum simulation outputs - interpreting and ranking results by clinical relevance after the quantum kernel has produced raw binding-energy estimates.
DFT	Density Functional Theory	A workhorse classical method for electronic-structure calculations. Used in Layer 2 for preprocessing - determining the active space (which electrons enter the quantum simulation, which remain classical) before the quantum kernel runs. Also serves as a benchmark in the results comparison: at C275F's active-space boundary, DFT-B3LYP misses chemical accuracy by 2.8 kcal/mol.
DMRG	Density Matrix Renormalization Group	A classical numerical method exceptionally accurate for one-dimensional or weakly entangled systems. Listed in Layer 2 alongside DFT as a classical pre/post-processing tool: provides classical baselines and treats fragments outside the quantum active space.
DSR	Design Science Research	The methodological framework anchoring this dissertation, after Hevner et al. (2004). DSR positions the work as the design and evaluation of an artifact (the platform) rather than as hypothesis-testing science. Phase 4 implements DSR-prescribed expert evaluation with an n=4–6 panel, OSF pre-registered, against four criteria (utility, novelty, rigor, feasibility).
EGFR	Epidermal Growth Factor Receptor	An oncogene with FDA-approved targeted therapies for NSCLC (osimertinib, erlotinib). Referenced in the Problem Statement to delineate the clinical gap: ~60% of NSCLC patients lack EGFR/ALK/KRAS-G12C mutations and therefore have no approved targeted therapy - the patient population this dissertation addresses.
EWf-SQD	Embedded Wave Function - Sample-based Quantum Diagonalization	A hybrid quantum-classical workflow announced by IBM and Cleveland Clinic in March 2026, demonstrated on the 303-atom Trp-cage miniprotein. Combines wave-function-based fragment embedding with SQD on each cluster. Recognized late in the dissertation cycle as an adjacent methodology with potential application to scaffold-perturbation mutations like C275F; its convergence guarantees are weaker than SqDRIFT's, which becomes a design consideration if the methods are hybridized.
FCI	Full Configuration Interaction	The exact solution to the electronic Schrödinger equation within a given basis set. Computationally infeasible beyond very small systems: $O(2^N)$ scaling reaches $\sim 10^{13}$ operations at $N=44$, which the poster cites as completely intractable. FCI defines the unreachable upper bound that quantum methods aim to approximate.
FDA	U.S. Food and Drug Administration	The federal agency regulating pharmaceuticals, medical devices, and clinical research in the United States. This dissertation's Layer 5 contribution operates within FDA's 21 CFR Part 11 framework (electronic records and signatures) - designing audit-trail architecture for probabilistic quantum simulation outputs.
FEP+	Free Energy Perturbation Plus (Schrödinger's commercial implementation)	A leading classical method for computing protein-ligand binding affinities. Cited in Layer 3 (benchmarking) as the industrial classical baseline against which quantum methods must be compared in terms of wall-clock time, accuracy, and active-space coverage.
GARD	Genomic-Adjusted Radiation Dose	A radiogenomic model predicting individual tumor radiosensitivity from gene-expression signatures. Mentioned in the Future Research Direction on the poster: post-doctoral work would extend the dissertation's quantum-derived mechanistic features into GARD-class radiotherapy decision support for TP53-mutant tumors. Out of dissertation scope.
GNN	Graph Neural Network	A neural-network architecture operating on graph-structured data - natural for molecules represented as atoms (nodes) and bonds (edges). DeepChem GNN is the AI rescoring component in Layer 2, interpreting raw quantum-simulation outputs in terms of drug-likeness and clinical relevance.
GPU	Graphics Processing Unit	A massively parallel processor originally designed for graphics, now central to scientific computing. NVQLink provides the GPU ↔ QPU interconnect at the heart of Layer 2; classical pre/post-processing (DFT, DMRG, GNN inference) runs on GPU clusters.
HIPAA	Health Insurance Portability and Accountability Act	U.S. federal law (1996) governing privacy of medical information. The dissertation works exclusively with de-identified single-case data (MI25-0349) to remain outside HIPAA's PHI requirements. HIPAA compliance for multi-patient analysis is reserved for post-doctoral work.
HPC	High-Performance Computing	Large-scale classical computing infrastructure (CPU/GPU clusters). The third pillar of the QC·AI·HPC architecture: HPC handles classical pre-processing (DFT, DMRG) and post-processing (AI rescoring) that surrounds the quantum kernel.
IND	Investigational New Drug	An FDA application (governed by 21 CFR Part 312) required before administering an experimental drug to humans. Out of dissertation scope by explicit decision: Part 312 is a regulatory pathway, while this dissertation's contribution is the upstream Part 11 audit infrastructure that any future IND submission relying on quantum-simulation evidence would require.
IRB	Institutional Review Board	The ethics committee approving research involving human subjects. The dissertation operates under an IRB scope limited to anonymized single-case reference (MI25-0349); multi-patient analysis is deferred to post-doctoral work.
IS	Information Systems	The disciplinary home of this dissertation. Positioning the work as IS - architecture, governance, audit-trail design - rather than as computational chemistry is what makes it appropriate for CGU CISAT's DTech program.
JW	Jordan-Wigner (fermion-to-qubit mapping)	One of three fermion-to-qubit mappings the Phase 3A pipeline implements and compares (alongside parity+two-qubit-reduction and Bravyi-Kitaev). The mapper choice is recorded in audit-trail field P2 (compilation lineage).
KEAP1	Kelch-like ECH-Associated Protein 1	One of the three KEAP1 target mutations (LOF, G333C, R320Q), classified Class B (quantum-advantaged). The Kelch-domain binding pocket — which mediates the Nrf2 interaction disrupted in cancer — spans a full active site of ~155 active electrons (~310 qubits under Jordan-Wigner encoding), well beyond the classical tractability ceiling (~18e). Designated fault-tolerant QPU target (~2030+); coordinate-verified from PDB structures 1U6D and 2FLU.

KRAS	Kirsten Rat Sarcoma viral oncogene homolog	An oncogene mutated in ~25% of NSCLC. KRAS G12C now has an FDA-approved targeted therapy (sotorasib), placing it among the druggable mutations referenced in the Problem Statement. Most other KRAS variants and the four mutations addressed in this dissertation remain non-druggable.
L2 / L3 (loops)	L2 and L3 loops of the p53 DNA-binding domain	Structural loops in the p53 DBD that stabilize Zn ²⁺ tetrahedral coordination via Cys176, His179, Cys238, Cys242. The loop-sheet-helix motif containing C275 is structurally adjacent. Relevant to active-space selection: the local 24-electron Phase 3A space and the full 44-electron Phase 3B space differ in how much of L2/L3 they include.
ML	Machine Learning	A subfield of AI focused on systems that learn patterns from data. The Nvidia Ising 3D CNN QEC decoder is an ML component; its provenance - model version, weights, calibration data - is captured in audit-trail field P9 (ML-decoder lineage).
MM	Molecular Mechanics	Classical force-field treatment of atomic motion. The classical half of QM/MM hybrid partitioning: the bulk of the protein is treated by MM while the chemically critical region around the mutation site receives quantum treatment.
MTD	Maximum Tolerated Dose	The highest drug dose that does not produce unacceptable toxicity. Mentioned in clinical-trial context (e.g., the rezatapopt PYNACLE Phase 1 endpoint). Out of dissertation scope; relevant only to clinical-translation discussions in the Future Research section.
NDA	New Drug Application	The FDA filing seeking approval to market a new drug. Out of dissertation scope; mentioned in connection with rezatapopt (PMV Pharma's NDA targeted for Q1 2027 for TP53 Y220C) to clarify why Y220C is no longer non-druggable.
NEJM	New England Journal of Medicine	A leading medical journal. Cited as the venue (February 2026) for the rezatapopt Phase 1 results in TP53 Y220C - the publication that confirms Y220C's transition from non-druggable to druggable, prompting its removal from the dissertation's mutation list.
NGS	Next-Generation Sequencing	High-throughput DNA sequencing technology used to identify mutations in patient tumors. The dissertation's anchor patient case MI25-0349 is sourced from NGS data: TP53 p.Cys275Phe identified as a Tier 1 pathogenic variant at 12.66% allele frequency (Emek Medical Center, June 2025).
NSCLC	Non-Small Cell Lung Cancer	The most common form of lung cancer (~85% of cases) and the leading cause of cancer mortality globally (~2.1 million new cases and ~1.53 million deaths annually; GLOBOCAN 2022). The clinical context of the dissertation: the four target mutations (TP53, KEAP1, CDKN2A, STK11) are common non-druggable drivers in NSCLC patients.
Nvidia	Nvidia Corporation	GPU manufacturer providing key Layer 2 components: CUDA-Q programming model, NVQLink GPU ↔ QPU interconnect, and Ising AI stack (VLM calibration + 3D CNN QEC decoder). Nvidia's emergence as cross-vendor infrastructure is what made the dissertation's earlier "cross-vendor abstraction" claim a commodity, prompting the narrowing to Part-11 governance as the defensible white space.
NVQLink	Nvidia's GPU ↔ QPU interconnect	A hardware-software fabric enabling tight integration between GPU clusters and quantum processors. Layer 2 component (commodity); together with CUDA-Q, it represents the vendor-spanning quantum-classical infrastructure the dissertation builds on top of rather than reinventing.
OSF	Open Science Framework	A platform for pre-registration of research designs and hypotheses. The dissertation's Phase 4 evaluation criteria are pre-registered on OSF before evaluation begins - the methodological commitment that ensures Layer 1 acceptance thresholds cannot be retrofitted to results.
P1- P9	Provenance fields 1 through 9 of the audit-trail schema	The dissertation's signature contribution to Information Systems: a nine-field provenance record that makes probabilistic quantum simulation outputs auditable under FDA 21 CFR Part 11. P1 circuit fingerprint, P2 compilation lineage, P3 device identifier and calibration epoch, P4 per-gate error budget snapshot, P5 shot count and raw outcome distribution, P6 error-mitigation lineage, P7 statistical estimator and confidence interval, P8 cryptographic seal, P9 ML-decoder provenance. Each field maps to a specific 21 CFR §11.10 requirement.
Part 11	21 CFR Part 11 (FDA Electronic Records & Signatures rule)	The federal regulation governing electronic records and signatures in FDA-regulated contexts. The dissertation's Layer 5 / §06.iii contribution: the first systematic framework applying Part 11 to probabilistic quantum-computed binding data. Part 11 (records) is in scope; Part 312 (IND submission) is explicitly out of scope.
Part 312	21 CFR Part 312 (FDA Investigational New Drug regulation)	The federal regulation governing IND applications. Explicitly excluded from the dissertation's novelty claim - Part 312 is a regulatory pathway, while the contribution sits upstream in the Part 11 records infrastructure that any future IND filing using quantum evidence would depend on.
PennyLane	PennyLane (Xanadu's quantum-machine-learning framework)	The simulator backend for Phase 3A. Currently live at 4 qubits (CAS(2e,2o)/STO-3G, AllSinglesDoubles UCCSD ansatz, 80 Adam steps) on Render cloud. The same PennyLane pipeline scales to the intermediate target (~20e/40q, ~17.6 TB state-vector) on NSF ACCESS / USC CARC HPC, enabling architecture validation at meaningful quantum-circuit scale without IBM Quantum Network access before Phase 3B hardware execution. The scientific ceiling target is ~24e/48q (~4.5 PB state-vector, edge of global classical capacity); the operational run uses ~20e/40q, which is well within multi-node HPC reach while still demonstrating two orders of magnitude of scale beyond the 4-qubit minimal prototype.
PHI	Protected Health Information	Patient-identifying medical data protected under HIPAA's eighteen-identifier rule. The dissertation works only with de-identified data (MI25-0349 is an internal study identifier with no link to patient identity), placing it outside PHI/HIPAA scope and simplifying compliance dramatically.
PMV	PMV Pharmaceuticals	The biotech company developing rezatapopt (PC14586) for TP53 Y220C. PMV's Phase 2 PYNACLE trial and planned Q1 2027 NDA are the reason Y220C transitioned from non-druggable to druggable and was removed from the dissertation's mutation list.
PYNACLE	Phase 1/2 trial of rezatapopt in TP53 Y220C-mutated solid tumors (NCT04585750)	The clinical trial demonstrating rezatapopt's efficacy in Y220C-mutated cancers (33% ORR across 8 tumor types as of August 2025). Cited as the evidence base for removing Y220C from the dissertation's non-druggable-mutation list.
QC	Quantum Computing	Computation exploiting quantum-mechanical phenomena (superposition, entanglement) to solve problems intractable for classical hardware. The first pillar of the QC-AI-HPC architecture; Layer 1 of the platform houses the quantum kernel (SqDRIFT on IBM Heron r3).
QCS	Quantum Cloud Services (IBM)	IBM's cloud platform for quantum-hardware access. Listed alongside Nvidia NVQLink in the bifurcation-aware orchestration claim: the platform's architecture must survive the IBM-QCS vs Nvidia-NVQLink vendor split that emerged during the proposal cycle.

QEC	Quantum Error Correction	The set of techniques for protecting quantum computations against noise by encoding logical qubits across many physical qubits with active error detection. The Nvidia Ising 3D CNN decoder is the QEC decoder in Layer 2; its operation generates the data captured in audit-trail field P9 (ML-decoder lineage).
QEM	Quantum Error Mitigation	Techniques (ZNE, probabilistic error cancellation, readout correction) that reduce - but do not eliminate - the impact of noise on quantum computations, without the qubit overhead of full QEC. Recorded in audit-trail field P6 (error-mitigation lineage). Distinguished from QEC: mitigation is post-processing; correction operates during execution.
QM	Quantum Mechanics	The physics framework underlying quantum chemistry and quantum computing. The quantum half of QM/MM hybrid partitioning: the chemically critical region around the mutation site receives QM treatment while the bulk of the protein is treated classically by MM.
QM/MM	Quantum Mechanics / Molecular Mechanics hybrid partitioning	A multi-scale modeling approach in which a small chemically critical region is treated quantum-mechanically while the surrounding environment is treated classically. The dissertation requires a novel QM/MM scheme tailored to TP53's Zn ²⁺ site for C275F - itself a research contribution within the broader pipeline.
QPU	Quantum Processing Unit	The hardware unit performing quantum computation, analogous to a CPU or GPU. IBM Heron r3 is the QPU targeted in Phase 3B. NVQLink provides the GPU ↔ QPU interconnect.
QV	Quantum Volume	An IBM-defined benchmark capturing combined qubit count, connectivity, and gate fidelity. The poster lists IBM Heron r3 at QV ≥ 2048 - establishing that the hardware regime of the IBM Research 2026 anchor (32 electrons, 100 qubits) is reproducible at Phase 3B scale.
RP2D	Recommended Phase 2 Dose	The drug dose level chosen from Phase 1 results to take into Phase 2 trials. Mentioned in clinical-trial context (rezatapopt PYNNACLE) and out of dissertation scope; relevant only to the rationale for removing Y220C from the mutation list.
SC150	Stabilization Concentration 150 (50% protein-stabilization endpoint)	A biochemical potency measure for protein-stabilizing drugs. Cited in the rezatapopt context (SC150 = 9 nM for Y220C binding) - the kind of experimental ground-truth value that, were Y220C retained as a positive control, the dissertation's quantum platform could be benchmarked against.
SQD	Sample-based Quantum Diagonalization	A family of hybrid quantum-classical algorithms where the quantum computer samples configurations and the classical computer diagonalizes a reduced Hamiltonian over those samples. SqDRIFT is one variant; EWF-SQD is another. Convergence in the SQD family is generally empirical rather than provable.
SqDRIFT	Sample-based quantum Diagonalization with stochastic Trotter (qDRIFT)	The quantum algorithm anchoring Layer 1 of the dissertation. Demonstrated by IBM in 2026 (IBM Research; arXiv:2603.08696 and the SqDRIFT method paper arXiv:2508.02578): a 32-electron active space of C ₁₃ Cl ₂ on 100 qubits of Heron, beyond the classical CASSCF ceiling of ~18 electrons. SqDRIFT inherits qDRIFT's provable convergence bounds - a strength the dissertation contrasts with EWF-SQD's empirical-only convergence.
STK11	Serine/Threonine Kinase 11 (also LKB1)	One of the four STK11/LKB1 target mutations (LKB1 L0F, F354L, D194N), classified Class A (quantum-necessary). Full active site: ~152 active electrons (~304 qubits under Jordan-Wigner encoding) — designated fault-tolerant QPU target (~2030+). STK11 F354L local site (~48e/~96q; AlphaFold Q15831) is a near-term exception executable on IBM Heron r3, only 2 qubits beyond the demonstrated 94-qubit ceiling (Merz et al. 2026, arXiv:2605.01138). PDB source: 2WTK.
TP53	Tumor Protein 53 (the p53 tumor suppressor gene)	The most frequently mutated gene in human cancers (~59% of solid tumors). The dissertation focuses on TP53 structural (DNA-interface) mutants - particularly C275F - that disrupt DNA-binding geometry without creating druggable pockets. TP53's Zn ²⁺ tetrahedral coordination site (Cys176/His179/Cys238/Cys242) requires ~22–28 active electrons, the paradigmatic case for SqDRIFT-grade simulation.
Trp-cage	Tryptophan-cage miniprotein	A 20-residue, 303-atom synthetic miniprotein. The substrate of the Cleveland Clinic / IBM EWF-SQD demonstration (March 2026) - the first protein-scale SQD-family simulation. Used in the dissertation's positioning to argue that EWF-SQD validates protein-scale viability without scooping the dissertation's clinical-mutation focus.
VLM	Vision-Language Model	A large multimodal AI model processing images and text jointly. Nvidia's Ising calibration component (35-billion-parameter VLM) autonomously brings up QPU calibration. Layer 2 commodity; its outputs feed the QEC decoder whose lineage is captured in P9.
VQE	Variational Quantum Eigensolver	A hybrid quantum-classical algorithm that uses a classical optimizer to minimize the expectation value of a quantum-prepared trial state. Phase 3A benchmarks VQE outputs against DFT and CCSD(T) reference values on the C275F local active site. The poster's results illustration shows VQE-SqDRIFT at -11.7 kcal/mol with 0.5 kcal/mol error - within chemical accuracy.
Y220C	TP53 p.Tyr220Cys (tyrosine-to-cysteine substitution at position 220)	A TP53 missense mutation creating a small drug-binding pocket. Originally listed among the dissertation's five target mutations but removed once rezatapopt (PMV Pharma's first-in-class p53 reactivator) entered Phase 2 with FDA Fast Track designation, reclassifying Y220C as druggable. The current target list is four mutations: C275F, KEAP1, CDKN2A, STK11.
ZNE	Zero-Noise Extrapolation	A quantum error-mitigation technique that runs circuits at varying noise levels and extrapolates to the zero-noise limit. Listed among the QEM techniques whose vendor-specific implementation differences (BPR problem P2) the orchestration platform must standardize. Recorded in audit-trail field P6.

Note on scope. This glossary is descriptive of the work as it stands at the proposal stage. Some terms (e.g., EWF-SQD, ZNE) appear in the dissertation only as comparison points or discussion items, while others (e.g., SqDRIFT, BQP, Part 11, P1–P9) are core to the contribution. The "Definition & Context" column makes this distinction explicit for each entry.

Living document. As the dissertation progresses through Phases 3A and 3B, additional terms (specific calibration data fields, hardware identifiers, evaluation metrics) will enter the artifact set. This glossary will be updated alongside each major revision of the proposal or platform.